

CYBERSECURITY MONITORING LAB

SIEM, XDR, IDS and SOAR-Based Security Monitoring using Wazuh and Suricata

Project Report

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ABSTRACT

Cyber threats are continuously evolving, making traditional security monitoring insufficient for modern organizations. Security Information and Event Management (SIEM), Extended Detection and Response (XDR), Intrusion Detection Systems (IDS), and Security Orchestration, Automation and Response (SOAR) technologies play a crucial role in identifying, analyzing, and responding to cyberattacks. This project presents the design and implementation of a cybersecurity monitoring laboratory using Wazuh SIEM, Suricata IDS, and Ubuntu Linux. The environment simulates real-world attacks from a Kali Linux attacker machine and demonstrates centralized log collection, threat detection, alert generation, attack visibility, and automated notification capabilities. The project also explores ARP spoofing attacks, network monitoring, active response concepts, and email-based alerting mechanisms. The implemented solution provides a cost-effective open-source security monitoring platform suitable for learning, research, and small-scale enterprise environments. The results demonstrate the effectiveness of combining multiple security technologies in a unified monitoring framework to detect and respond to modern cyber threats in real time.

Keywords: SIEM, XDR, Wazuh, Suricata, SOAR, IDS, Cybersecurity Monitoring, ARP Spoofing, Threat Detection, Security Operations Center.

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CHAPTER 1: INTRODUCTION

1.1 Overview of Cybersecurity Monitoring

In the digital age, organizations face an ever-increasing volume and sophistication of cyber threats. From nation-state actors conducting espionage to ransomware groups targeting critical infrastructure, the threat landscape has grown dramatically in both complexity and scale. Security monitoring — the continuous observation and analysis of an organization's information systems — has become a foundational pillar of modern cybersecurity strategy.

Cybersecurity monitoring involves collecting, aggregating, and analyzing logs, events, and alerts from disparate sources across an IT environment. These sources include endpoint devices, servers, network appliances, applications, databases, and cloud services. Without centralized visibility, security analysts struggle to detect threats, correlate events, and respond effectively. As organizations increasingly adopt hybrid and cloud-based infrastructures, maintaining comprehensive visibility across distributed environments has become both more challenging and more critical.

Security Information and Event Management (SIEM) platforms emerged as a solution to this challenge by providing centralized log collection, correlation, and alerting capabilities. Modern SIEM solutions have evolved beyond simple log management to incorporate advanced analytics, machine learning, threat intelligence integration, and automated response workflows. These capabilities enable security teams to identify anomalous behavior, detect sophisticated attack patterns, and reduce the time required to investigate incidents.

Another significant driver for security monitoring is regulatory compliance. Standards and frameworks such as GDPR, HIPAA, PCI-DSS, ISO 27001, and NIST require organizations to maintain audit trails, monitor security events, and demonstrate effective incident detection and response capabilities. SIEM platforms help organizations meet these requirements by providing centralized logging, reporting, and evidence collection.

Network-based monitoring solutions such as Intrusion Detection Systems (IDS) play a crucial role in identifying malicious traffic and unauthorized activities. By inspecting network packets and comparing them against known attack signatures or behavioral patterns, IDS technologies provide an additional layer of defense that complements endpoint and host-based monitoring.

The integration of threat intelligence feeds further enhances monitoring effectiveness by enabling organizations to identify indicators of compromise (IOCs), malicious IP addresses, suspicious domains, and emerging attack techniques. This proactive approach helps security teams stay ahead of evolving threats and improve their overall security posture.

Security Orchestration, Automation, and Response (SOAR) platforms extend monitoring capabilities by automating repetitive tasks such as alert triage, incident enrichment, and response actions. Automation reduces analyst workload, minimizes response times, and allows security teams to focus on high-priority threats and strategic security initiatives.

When combined with Intrusion Detection Systems (IDS) for network-level monitoring and Security Orchestration, Automation and Response (SOAR) for automated incident handling, organizations can build a robust Security Operations Center (SOC) capable of detecting and responding to threats at scale. Effective SOC operations rely on real-time visibility, rapid incident response, continuous threat hunting, and comprehensive security analytics to minimize the impact of cyberattacks.

This project implements such an integrated monitoring environment using Wazuh — an open-source SIEM and Extended Detection and Response (XDR) platform — and Suricata, a high-performance network IDS/IPS engine. The laboratory environment simulates real-world attack scenarios, including reconnaissance, exploitation, malware activity, and unauthorized access attempts, to validate detection capabilities and explore automated response mechanisms. Through this implementation, the project demonstrates how open-source security technologies can be leveraged to build a cost-effective, scalable, and efficient cybersecurity monitoring solution suitable for modern enterprise environments.

1.2 Problem Statement

Traditional security monitoring approaches suffer from critical deficiencies that leave organizations vulnerable to modern cyber threats. Many organizations rely on siloed security tools that operate independently without sharing context or correlating events. This lack of integration results in alert fatigue, where analysts are overwhelmed by large volumes of uncontextualized notifications, causing critical threats to potentially go unnoticed amidst the noise.

Furthermore, network-level threats such as port scanning, ARP spoofing, Man-in-the-Middle (MITM) attacks, brute-force attempts, and lateral movement activities often go undetected without dedicated Intrusion Detection Systems (IDS). The absence of real-time threat detection and automated response capabilities significantly extends the time between initial compromise and detection—a metric known as Mean Time to Detect (MTTD)—which directly impacts the severity, financial cost, and operational disruption caused by security incidents.

The rapid adoption of cloud computing, remote work environments, and Bring Your Own Device (BYOD) policies has further expanded the attack surface of modern organizations. Security teams must monitor a growing number of endpoints, cloud workloads, and network connections, making it increasingly difficult to maintain visibility across the entire infrastructure using traditional monitoring methods.

Another major challenge is the increasing sophistication of cyberattacks. Modern adversaries employ advanced persistent threats (APTs), fileless malware, encrypted command-and-control communications, and multi-stage attack techniques designed to evade conventional security controls. Traditional monitoring systems often lack the behavioral analytics and correlation capabilities required to identify such complex attack patterns.

Organizations also face a shortage of skilled cybersecurity professionals capable of continuously monitoring, investigating, and responding to security incidents. Security analysts frequently spend substantial time manually reviewing logs, validating alerts, and correlating events from multiple sources, which can delay incident response and increase the risk of successful attacks.

Additionally, many organizations struggle to implement effective incident response processes due to the lack of automation and centralized management. Without automated workflows, containment and remediation actions may be delayed, allowing attackers additional time to escalate privileges, exfiltrate sensitive data, or disrupt critical services.

Resource constraints present another significant obstacle. Enterprise-grade security solutions are often prohibitively expensive for small and medium-sized enterprises (SMEs), educational institutions, and research laboratories. Licensing costs, infrastructure requirements, and specialized personnel expenses create barriers to implementing comprehensive security monitoring solutions. Consequently, there is a growing need for cost-effective, scalable, and open-source alternatives that can provide enterprise-level visibility and threat detection capabilities without substantial financial investment.

This project addresses these challenges by designing and implementing an integrated, open-source cybersecurity monitoring laboratory that combines centralized log management, network intrusion detection, automated alert generation, threat correlation, and incident response capabilities. By leveraging Wazuh and Suricata, the proposed solution demonstrates how organizations can achieve enhanced visibility, real-time threat detection, reduced response times, and improved overall security posture while maintaining affordability and scalability.

1.3 Objectives of the Project

The primary objective of this project is to design, implement, and validate a comprehensive cybersecurity monitoring laboratory using open-source tools. The system aims to provide security analysts, researchers, and students with a practical platform for learning threat detection, incident analysis, security monitoring, and Security Operations Center (SOC) workflows. The project focuses on demonstrating how integrated security technologies can improve organizational visibility, accelerate threat detection, and enhance incident response capabilities in modern IT environments. The specific objectives of the project are as follows:

- To implement a SIEM platform using Wazuh for centralized log collection, event correlation, and threat detection.
- To integrate Suricata IDS for network traffic analysis and signature-based intrusion detection.
- To simulate real-world cyber attacks using Kali Linux, including Nmap reconnaissance and ARP spoofing.
- To demonstrate Extended Detection and Response (XDR) capabilities through unified endpoint and network monitoring.
- To generate and visualize security alerts through the Wazuh Dashboard for incident investigation.
- To implement email-based alert notifications as a foundational SOAR capability.
- To explore automated response mechanisms for critical security events.
- To analyze the performance and effectiveness of the monitoring solution in detecting simulated attacks.

1.4 Scope of the Project

The scope of this project encompasses the complete design, implementation, configuration, and validation of a virtual cybersecurity monitoring laboratory using open-source security technologies. The project

aims to create a realistic yet controlled environment for studying cybersecurity monitoring, threat detection, incident analysis, and security operations workflows. By utilizing virtualization technology through VirtualBox, the laboratory provides a safe platform for conducting attack simulations and evaluating detection mechanisms without impacting production systems or organizational networks.

The project includes the deployment and integration of multiple security components, including the Wazuh SIEM and XDR platform, Suricata Intrusion Detection System (IDS), OpenSearch for data storage and indexing, and the Wazuh Dashboard for visualization and security analytics. These components work together to provide centralized monitoring, log management, event correlation, network traffic analysis, and alert generation capabilities.

The implemented environment monitors Ubuntu Linux endpoints through the Wazuh agent, collecting system logs, security events, authentication records, file integrity monitoring data, and endpoint activity information. Simultaneously, Suricata performs network traffic inspection to identify suspicious communications, known attack signatures, and anomalous network behavior. The integration of host-based and network-based monitoring demonstrates a comprehensive security monitoring architecture aligned with modern Security Operations Center (SOC) practices.

The scope further includes the simulation and detection of common cyberattack scenarios using Kali Linux as an attacker machine. These scenarios include network reconnaissance using Nmap, port scanning, service enumeration, ARP spoofing attacks, unauthorized access attempts, and other network-based attack techniques. The project evaluates how effectively the monitoring infrastructure detects, correlates, and reports these activities in real time.

Additionally, the project covers alert management and notification mechanisms through the Wazuh Dashboard and email-based alerting workflows. Security events are visualized using dashboards and reports, enabling analysts to investigate incidents, identify attack patterns, and understand system behavior. Basic automation features are explored to demonstrate the foundational concepts of Security Orchestration, Automation, and Response (SOAR).

The project also includes performance evaluation of the monitoring solution by analyzing detection accuracy, alert generation efficiency, system visibility, and response capabilities under simulated attack conditions. The effectiveness of integrating endpoint monitoring with network intrusion detection is assessed to demonstrate the benefits of a unified security monitoring approach.

However, the scope of the project is limited to a laboratory environment intended primarily for educational, research, and demonstration purposes. The implementation does not include enterprise-scale deployment features such as high availability, distributed architectures, multi-site monitoring, advanced threat intelligence integration, large-scale log ingestion, or comprehensive compliance management. The project focuses on foundational cybersecurity monitoring concepts rather than full-scale production deployment.

Furthermore, the project does not replace commercial security solutions and should not be considered a complete defense strategy for enterprise environments. Real-world implementations would require

additional security hardening, infrastructure redundancy, access control mechanisms, scalability planning, continuous tuning of detection rules, compliance alignment, and integration with organizational security policies and incident response procedures.

Despite these limitations, the project provides a scalable framework that can be extended in future work to incorporate advanced SOAR capabilities, machine learning-based anomaly detection, cloud security monitoring, threat intelligence feeds, vulnerability management systems, and additional security tools. As such, it serves as both a practical learning platform and a foundation for future cybersecurity research and development.

1.5 Motivation

The motivation for this project stems from the growing importance of cybersecurity in an increasingly interconnected and technology-driven world. Organizations across all sectors—including government agencies, financial institutions, healthcare providers, educational organizations, and private enterprises—depend heavily on digital infrastructure to support their operations. As this dependence continues to grow, cyber threats have become more frequent, sophisticated, and damaging. High-profile data breaches, ransomware attacks targeting critical infrastructure, supply chain compromises, and advanced persistent threats (APTs) have demonstrated the significant financial, operational, and reputational consequences of inadequate cybersecurity measures.

One of the most critical challenges faced by organizations today is the ability to detect and respond to security incidents in a timely manner. Attackers often remain undetected within networks for extended periods, allowing them to gather intelligence, escalate privileges, exfiltrate sensitive data, or disrupt critical services. Effective security monitoring is therefore essential for reducing attack dwell time, improving situational awareness, and enabling rapid incident response. This project is motivated by the need to better understand and demonstrate how modern security monitoring solutions can help organizations identify and mitigate cyber threats before they cause significant damage.

Another key motivation is the increasing demand for Security Operations Center (SOC) capabilities. Modern SOC's rely on centralized monitoring, threat detection, event correlation, and automated response mechanisms to protect organizational assets. However, many organizations—particularly small and medium-sized enterprises (SMEs), startups, and educational institutions—lack the financial resources necessary to deploy and maintain expensive commercial SIEM and XDR solutions. Open-source security platforms provide an attractive alternative by offering many of the same capabilities at significantly lower costs.

Open-source security tools such as Wazuh and Suricata have matured considerably over the past decade and now provide advanced functionalities including log management, endpoint monitoring, network intrusion detection, threat intelligence integration, file integrity monitoring, vulnerability detection, and security analytics. These tools enable organizations to build effective security monitoring infrastructures without incurring substantial licensing expenses. This project seeks to demonstrate the practical viability,

effectiveness, and scalability of open-source security technologies as cost-effective alternatives to proprietary solutions.

The project is also motivated by the growing importance of cybersecurity education and workforce development. The global shortage of skilled cybersecurity professionals has created a pressing need for practical training environments that allow students and researchers to gain hands-on experience with industry-relevant tools and technologies. While many academic programs provide theoretical knowledge of cybersecurity concepts, students often have limited opportunities to work with real-world monitoring systems, investigate security alerts, analyze attack behavior, and perform incident response activities.

By creating a complete and reproducible cybersecurity monitoring laboratory, this project bridges the gap between theory and practice. The laboratory environment enables learners to observe attack techniques, generate security events, investigate alerts, and understand the operational workflows used by security analysts in professional SOC environments. Such practical exposure helps develop critical analytical, technical, and problem-solving skills required in modern cybersecurity careers.

Furthermore, this project serves as a platform for experimentation and research. The laboratory can be extended to evaluate new detection rules, test emerging attack techniques, explore automation strategies, and assess the effectiveness of different monitoring approaches. As cyber threats continue to evolve, having a flexible and adaptable research environment becomes increasingly valuable for advancing cybersecurity knowledge and innovation.

Ultimately, the motivation behind this project is to contribute to improved cybersecurity awareness, promote the adoption of accessible security technologies, support practical cybersecurity education, and demonstrate how integrated monitoring solutions can strengthen organizational defenses against modern cyber threats.

1.6 Organization of the Report

This report is organized into sixteen chapters to provide a comprehensive understanding of the project design, implementation, and results.

- Chapter 1: Introduction — provides an overview of cybersecurity monitoring, defines the problem statement, outlines objectives and scope, and describes the organization of the report.
- Chapter 2: Literature Survey — reviews existing research on SIEM, XDR, IDS, and SOAR technologies, examining key platforms and methodologies.
- Chapter 3: Technologies Used — details the technical specifications and capabilities of Wazuh, Suricata, OpenSearch, Kali Linux, and Ubuntu.
- Chapter 4: System Architecture — presents the overall system design, component interactions, and data flow within the monitoring laboratory.
- Chapter 5: Environment Setup — describes the hardware, software, and network configuration requirements for the laboratory environment.
- Chapter 6: Wazuh Installation and Configuration — covers the step-by-step installation and configuration of the Wazuh SIEM platform.

- Chapter 7: Suricata IDS Integration — explains the installation, rule configuration, and Wazuh integration of the Suricata IDS engine.
- Chapter 8: Attack Simulation — documents the Nmap scan simulation, detection methodology, and alert analysis.
- Chapter 9: ARP Spoofing Attack Simulation — presents the ARP spoofing attack scenario, detection strategy, and security implications.
- Chapter 10: XDR Capabilities — explores the extended detection and response features provided by the integrated platform.
- Chapter 11: Email-Based SOAR Implementation — describes the automated notification workflow and alert email configuration.
- Chapter 12: Results and Analysis — presents detection results, dashboard screenshots, and performance analysis.
- Chapter 13: Advantages — outlines the key benefits of the implemented monitoring solution.
- Chapter 14: Limitations — acknowledges the current constraints and areas for improvement.
- Chapter 15: Future Enhancements — suggests potential improvements and extensions to the current implementation.
- Chapter 16: Conclusion — summarizes the project findings and contributions.

CHAPTER 2: LITERATURE SURVEY

2.1 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) represents the convergence of two earlier security disciplines: Security Information Management (SIM), focused on long-term log storage and analysis, and Security Event Management (SEM), focused on real-time monitoring and alerting. The term SIEM was first coined by Gartner analysts Mark Nicolett and Amrit Williams in 2005, and the technology has since evolved significantly to become a cornerstone of modern cybersecurity operations.

Early SIEM systems functioned primarily as centralized log aggregators, collecting security logs from servers, workstations, firewalls, routers, applications, and other infrastructure components. Their primary purpose was to support compliance reporting, auditing requirements, and post-incident forensic investigations. While these first-generation systems improved visibility by consolidating logs into a single repository, they provided limited real-time threat detection capabilities and required extensive manual analysis by security personnel.

As cyber threats became increasingly sophisticated and organizations generated larger volumes of security data, the limitations of basic log management solutions became evident. Second-generation SIEM platforms introduced event correlation technologies that enabled analysts to identify relationships between seemingly unrelated events occurring across different systems. Correlation rules could automatically detect attack patterns, suspicious activities, and policy violations by analyzing events from multiple sources simultaneously. This represented a significant advancement in threat detection; however, these systems often generated large numbers of false positives and required continuous rule tuning and maintenance.

The rapid growth of enterprise networks, cloud computing, mobile devices, and Internet of Things (IoT) technologies further increased the complexity of security monitoring. Organizations began collecting millions of security events daily, making manual analysis impractical. To address these challenges, modern SIEM solutions incorporated advanced analytics, big data processing, and automated detection techniques capable of handling large-scale security data environments.

Contemporary SIEM platforms now integrate User and Entity Behavior Analytics (UEBA), machine learning algorithms, threat intelligence feeds, and risk-based alerting mechanisms. These capabilities enable the detection of anomalous behavior, insider threats, credential misuse, account compromise, and advanced persistent threats (APTs) that may bypass traditional signature-based detection methods. Rather than relying solely on predefined rules, modern SIEM systems can identify deviations from normal user and system behavior, significantly improving threat detection accuracy.

Another major evolution in SIEM technology is its integration with Security Operations Centers (SOCs). SIEM platforms serve as the central nervous system of SOC environments by providing a unified view of

security events across endpoints, networks, cloud platforms, and applications. Security analysts utilize SIEM dashboards to monitor alerts, investigate incidents, perform threat hunting activities, and coordinate incident response efforts. The centralized visibility provided by SIEM solutions helps organizations reduce Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR), two critical metrics for effective cybersecurity operations.

In recent years, SIEM technology has increasingly converged with Extended Detection and Response (XDR) and Security Orchestration, Automation, and Response (SOAR) capabilities. XDR extends visibility across multiple security layers, including endpoints, networks, cloud environments, and identity systems, enabling more comprehensive threat detection and investigation. SOAR capabilities enhance SIEM platforms by automating repetitive tasks such as alert enrichment, incident triage, ticket creation, and response actions, thereby improving operational efficiency and reducing analyst workload.

SIEM systems also play a vital role in helping organizations meet regulatory and compliance requirements. Frameworks such as GDPR, HIPAA, PCI-DSS, ISO 27001, NIST Cybersecurity Framework, and SOC 2 require organizations to maintain audit logs, monitor security events, and demonstrate effective incident detection and response procedures. SIEM platforms facilitate compliance by providing centralized log retention, reporting capabilities, and audit trails necessary for regulatory assessments.

Despite their significant advancements, modern SIEM platforms continue to face challenges. Security teams must manage increasing volumes of data, address alert fatigue, reduce false positives, and adapt to constantly evolving attack techniques. Effective SIEM deployment requires proper configuration, correlation rule tuning, threat intelligence integration, and continuous monitoring to maximize detection accuracy and operational effectiveness.

Open-source platforms such as Wazuh have democratized access to SIEM technology by providing enterprise-grade security monitoring capabilities without the high licensing costs associated with many commercial solutions. Wazuh integrates log management, endpoint security monitoring, file integrity monitoring, vulnerability assessment, compliance monitoring, and threat detection into a unified platform. By leveraging open-source technologies, organizations, educational institutions, and researchers can implement comprehensive security monitoring infrastructures while maintaining flexibility, transparency, and cost efficiency.

As cyber threats continue to evolve, SIEM technology remains a critical component of modern cybersecurity strategies. The continued integration of artificial intelligence, automation, cloud-native architectures, and advanced analytics is expected to further enhance the effectiveness of SIEM platforms in detecting, investigating, and responding to increasingly sophisticated cyber threats.

2.2 Extended Detection and Response (XDR)

Extended Detection and Response (XDR) represents the next evolution in security monitoring technology, extending detection and response capabilities beyond the endpoint to encompass network, cloud, identity, email, and application layers. The concept emerged as a response to the increasing complexity of modern

cyber threats, which often traverse multiple systems and security domains during an attack lifecycle. The term was introduced by Palo Alto Networks in 2018 and has since gained widespread adoption across the cybersecurity industry as organizations seek more comprehensive and integrated threat detection solutions.

Traditional Endpoint Detection and Response (EDR) solutions focused primarily on endpoint devices such as workstations, servers, and laptops. These systems provided detailed visibility into endpoint activities including process execution, file system modifications, registry changes, user actions, and network connections. While EDR significantly improved endpoint security visibility, it operated largely in isolation and lacked the broader contextual awareness required to understand complex attacks spanning multiple systems and network layers.

As cyberattacks became increasingly sophisticated, attackers began employing multi-stage attack techniques that combined phishing, credential theft, lateral movement, command-and-control communications, privilege escalation, and data exfiltration. Detecting these attacks required correlating information from multiple security sources rather than relying solely on endpoint telemetry. This need led to the development of XDR as a unified security monitoring approach.

XDR addresses these limitations by integrating and correlating security data from multiple sources, including endpoint agents, network intrusion detection systems, cloud environments, email security platforms, identity and access management systems, firewalls, and threat intelligence feeds. By consolidating security telemetry into a centralized platform, XDR provides security teams with a more complete understanding of attack activities and enables faster identification of complex threats.

One of the key advantages of XDR is its ability to perform cross-domain correlation. Instead of analyzing events in isolation, XDR platforms connect related activities occurring across different security layers. For example, a phishing email, a suspicious endpoint process, unusual network communications, and unauthorized account activity can be correlated into a single attack narrative. This capability significantly improves detection accuracy while reducing alert fatigue caused by fragmented and disconnected security alerts.

Another important feature of XDR is enhanced incident investigation. Security analysts can trace attack progression across endpoints, networks, and cloud services from a single interface, eliminating the need to manually gather information from multiple tools. This centralized visibility improves operational efficiency and helps organizations reduce both Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).

Modern XDR platforms also incorporate automation and response capabilities. Automated workflows can isolate compromised endpoints, block malicious IP addresses, terminate suspicious processes, disable compromised accounts, and generate incident tickets based on predefined response rules. These capabilities reduce the burden on security analysts and enable faster containment of security incidents.

XDR plays a crucial role within Security Operations Centers (SOCs), where analysts must continuously monitor large volumes of security events. By consolidating data from diverse security controls into a

unified platform, XDR simplifies security operations and enables analysts to focus on high-priority threats rather than manually correlating alerts from multiple systems. This integration improves threat visibility and strengthens overall organizational security posture.

Wazuh exemplifies an open-source implementation of XDR principles by combining endpoint monitoring, log analysis, file integrity monitoring, vulnerability detection, compliance monitoring, and security analytics within a single platform. Through integration with Suricata IDS, Wazuh extends visibility beyond endpoints to include network traffic monitoring and intrusion detection. This integration enables correlation of endpoint events with network-based threats, providing security analysts with a comprehensive view of attack campaigns that span multiple hosts and network segments.

In the context of this project, the implementation of Wazuh and Suricata demonstrates how XDR concepts can be applied within a practical cybersecurity monitoring environment. Endpoint activities generated on monitored systems can be correlated with network-level events detected by Suricata, allowing the monitoring platform to identify attack patterns more effectively than either technology could independently. This unified approach enhances threat detection accuracy, improves incident investigation capabilities, and provides valuable insights into attacker behavior.

As cybersecurity environments continue to evolve, XDR is expected to become an increasingly important component of enterprise security architectures. Future XDR platforms will likely incorporate artificial intelligence, machine learning, behavioral analytics, cloud-native monitoring, and advanced automation technologies to further improve threat detection, investigation, and response capabilities. Consequently, XDR represents a significant advancement in cybersecurity operations by providing a holistic and integrated approach to security monitoring across modern digital infrastructures.

2.3 Intrusion Detection Systems

Intrusion Detection Systems (IDS) are security technologies designed to detect unauthorized access attempts, policy violations, malicious activities, and potential security threats within computer networks and information systems. IDS solutions play a critical role in modern cybersecurity architectures by providing continuous monitoring and alerting capabilities that enable organizations to identify and respond to security incidents before significant damage occurs. As cyber threats continue to evolve in sophistication and frequency, IDS technologies have become essential components of enterprise security infrastructures and Security Operations Centers (SOCs).

IDS technology can be broadly classified into two primary categories: Network-based Intrusion Detection Systems (NIDS) and Host-based Intrusion Detection Systems (HIDS). While both serve the common purpose of threat detection, they operate at different layers of the computing environment and provide complementary security capabilities.

Network-based Intrusion Detection Systems (NIDS) monitor network traffic at strategic locations within the infrastructure, such as network gateways, switches, routers, or network taps. By inspecting packets traversing the network, NIDS solutions can identify malicious activities including port scans, denial-of-service attacks, malware communications, reconnaissance attempts, protocol violations, and unauthorized

access attempts. Network-based monitoring provides visibility into communications between systems and enables detection of threats that may not be observable from individual endpoints.

Suricata, utilized in this project, is a high-performance open-source NIDS and Intrusion Prevention System (IPS) developed by the Open Information Security Foundation (OISF). Suricata is capable of multi-threaded packet processing, deep packet inspection, protocol analysis, file extraction, and application-layer monitoring. Unlike many traditional IDS solutions, Suricata can efficiently utilize modern multi-core processors, allowing it to analyze high volumes of network traffic with improved performance. Its compatibility with the Emerging Threats rule set enables detection of thousands of known attack signatures, malware indicators, exploit attempts, command-and-control communications, and suspicious network behaviors.

Host-based Intrusion Detection Systems (HIDS), in contrast, operate directly on individual hosts such as servers, workstations, and virtual machines. These systems monitor local activities including system calls, authentication events, file modifications, process executions, registry changes, user actions, and configuration alterations. HIDS solutions provide detailed visibility into endpoint behavior and can detect attacks that may not generate observable network indicators.

Wazuh incorporates robust HIDS capabilities through its lightweight agent architecture. The Wazuh agent continuously monitors endpoint activities and collects security-relevant information from operating systems and applications. Its features include file integrity monitoring (FIM), log analysis, rootkit detection, vulnerability assessment, security configuration assessment, and malware detection. These capabilities allow organizations to identify unauthorized changes, suspicious processes, privilege escalation attempts, and other indicators of compromise occurring at the host level.

Intrusion detection systems generally employ two primary detection methodologies: signature-based detection and anomaly-based detection. Signature-based detection relies on predefined patterns or rules representing known attack techniques. When observed activity matches a known signature, an alert is generated. This approach is highly effective against previously identified threats but may struggle to detect new or unknown attack methods. Anomaly-based detection, on the other hand, establishes a baseline of normal behavior and identifies deviations that may indicate malicious activity. While anomaly detection can identify novel threats, it often requires careful tuning to minimize false positives.

Another important distinction within intrusion detection technology is the difference between IDS and Intrusion Prevention Systems (IPS). IDS solutions are primarily passive monitoring tools that generate alerts when suspicious activities are detected. IPS solutions extend this functionality by actively blocking or mitigating malicious traffic in real time. Suricata supports both IDS and IPS operational modes, enabling organizations to choose between passive monitoring and active threat prevention based on their security requirements.

Despite their effectiveness, intrusion detection systems face several challenges. High volumes of network traffic can generate large numbers of alerts, potentially overwhelming security analysts. Signature-based systems require frequent rule updates to remain effective against emerging threats, while anomaly-based systems may produce false positives if behavioral baselines are not accurately established. Encrypted

network communications can also limit the visibility of network-based IDS solutions, making threat detection more difficult.

To address these limitations, modern cybersecurity strategies often employ a layered defense approach that combines both NIDS and HIDS technologies. Network-based monitoring provides visibility into communications and attack attempts traversing the infrastructure, while host-based monitoring provides detailed insight into endpoint activities and system-level events. The integration of these technologies enables more accurate threat detection and provides security analysts with richer contextual information during incident investigations.

In this project, the combination of Suricata and Wazuh demonstrates the effectiveness of a hybrid intrusion detection architecture. Suricata monitors network traffic and identifies malicious communications, reconnaissance activities, and attack signatures, while Wazuh provides endpoint visibility through host-based monitoring and log analysis. The integration of network and host-based security data enables comprehensive threat detection, improved event correlation, and enhanced incident response capabilities. This layered monitoring approach reflects modern cybersecurity best practices and forms a critical component of the project's cybersecurity monitoring laboratory.

As cyber threats continue to evolve, intrusion detection systems remain fundamental tools for protecting organizational assets. Advances in machine learning, behavioral analytics, threat intelligence integration, and automated response technologies are expected to further enhance IDS effectiveness, enabling organizations to detect and respond to increasingly sophisticated threats in real time.

2.4 Security Orchestration, Automation and Response (SOAR)

Security Orchestration, Automation, and Response (SOAR) platforms were developed to address the growing operational challenges faced by modern Security Operations Centers (SOCs). As organizations expanded their digital infrastructures, the volume of security alerts generated by SIEM systems, intrusion detection systems, endpoint security tools, cloud platforms, and other security technologies increased dramatically. At the same time, the global shortage of skilled cybersecurity professionals created significant pressure on security teams to manage increasing workloads with limited resources. SOAR emerged as a solution to improve operational efficiency, reduce manual effort, and accelerate incident response processes through automation and orchestration.

The concept of SOAR combines three key functions: security orchestration, security automation, and security response management. Together, these capabilities enable organizations to streamline security operations, improve consistency in incident handling, and reduce the time required to detect, investigate, and mitigate cyber threats.

Security orchestration refers to the integration and coordination of multiple security tools, technologies, and data sources within a unified workflow. Modern security environments often consist of numerous independent solutions, including SIEM platforms, firewalls, intrusion detection systems, endpoint protection tools, threat intelligence platforms, vulnerability scanners, and ticketing systems. Orchestration enables these technologies to communicate and share information through application programming

interfaces (APIs), allowing security teams to create coordinated workflows that span multiple security controls.

Security automation focuses on reducing the need for manual intervention in repetitive and time-consuming security tasks. Common automation activities include alert enrichment, log collection, threat intelligence lookups, malware analysis, user notification, ticket generation, and evidence gathering. By automating routine processes, organizations can reduce analyst workload, minimize human error, and ensure consistent execution of response procedures. Automation also enables security teams to respond more quickly to incidents, particularly during periods of high alert volume.

Incident response management is the third core component of SOAR platforms. These systems provide centralized case management capabilities that allow security teams to track, document, prioritize, and investigate incidents throughout their lifecycle. Incident response workflows help ensure that security events are handled systematically and that all relevant information is recorded for future analysis, auditing, and compliance purposes.

One of the most significant benefits of SOAR technology is its ability to reduce Mean Time to Respond (MTTR). Traditional incident response processes often require analysts to manually investigate alerts, gather information from multiple sources, and execute response actions across different systems. SOAR platforms automate many of these steps, allowing organizations to contain threats more rapidly and minimize the impact of security incidents. Faster response times are particularly important when dealing with ransomware attacks, data breaches, insider threats, and other time-sensitive security events.

SOAR platforms also help address the problem of alert fatigue, which occurs when analysts are overwhelmed by large numbers of security notifications. By automatically prioritizing alerts, enriching contextual information, and filtering low-priority events, SOAR solutions enable analysts to focus on incidents that present the greatest risk to the organization. This improves both operational efficiency and overall security effectiveness.

Modern SOAR implementations frequently support automated response actions such as blocking malicious IP addresses, isolating compromised endpoints, disabling user accounts, updating firewall rules, terminating suspicious processes, and notifying incident response teams. These actions can be executed automatically based on predefined rules or triggered manually following analyst approval, depending on organizational policies and risk tolerance.

Despite their advantages, SOAR platforms also present certain challenges. Developing effective automation workflows requires careful planning, extensive testing, and continuous maintenance. Incorrectly configured automated responses may disrupt legitimate business activities or generate unintended consequences. As a result, organizations often begin with low-risk automation tasks before gradually implementing more advanced response capabilities.

In recent years, SOAR technologies have increasingly converged with SIEM and Extended Detection and Response (XDR) platforms. Many modern security solutions now incorporate orchestration and automation features directly within their platforms, creating more integrated security operations

environments. This convergence enables organizations to detect threats, investigate incidents, and initiate response actions from a single management interface.

Within the context of this project, Wazuh provides foundational SOAR functionality through its active response framework and alerting mechanisms. Wazuh can automatically execute predefined actions when specific security events are detected, such as blocking malicious IP addresses, terminating suspicious connections, or triggering custom scripts. Additionally, the email notification system provides automated alert delivery to administrators, ensuring timely awareness of security incidents.

Although the implemented solution does not constitute a fully featured enterprise SOAR platform, it successfully demonstrates the fundamental principles of orchestration, automation, and incident response. The integration of Wazuh and Suricata allows security events to be detected, correlated, and acted upon through automated workflows, illustrating how security automation can improve operational efficiency and response effectiveness.

Furthermore, the laboratory environment created in this project provides a foundation for future SOAR enhancements. Additional capabilities such as threat intelligence integration, automated ticket creation, workflow orchestration, chatbot-assisted incident handling, and advanced response playbooks can be incorporated to create a more comprehensive automation framework. These future developments would further align the system with modern SOC practices and emerging trends in cybersecurity operations.

As cybersecurity threats continue to increase in volume and complexity, SOAR technology is expected to play an increasingly important role in security operations. By combining orchestration, automation, and response management, SOAR platforms enable organizations to improve security efficiency, reduce response times, and strengthen their overall cyber resilience in an increasingly challenging threat landscape.

2.5 Open Source Security Platforms

The open-source security ecosystem has matured significantly over the past decade, with several platforms now offering enterprise-grade capabilities without licensing costs. This evolution has been driven by active community contributions, increased cybersecurity awareness, and the growing demand for cost-effective security solutions. Open-source security tools are now widely adopted in academic environments, research laboratories, startups, and even large enterprises as part of hybrid security architectures.

Wazuh, originally a fork of OSSEC, has evolved into a comprehensive security monitoring platform that integrates SIEM, XDR, and compliance management capabilities into a unified solution. It provides functionalities such as log analysis, file integrity monitoring, vulnerability detection, rootkit detection, configuration assessment, and real-time alerting. Its integration with OpenSearch enables scalable data storage, indexing, and advanced search capabilities, allowing security teams to efficiently analyze large volumes of security events. Additionally, the Wazuh Dashboard offers intuitive visualization features that support incident investigation, threat hunting, and compliance reporting.

Suricata, developed by the Open Information Security Foundation (OISF), has established itself as a leading open-source network intrusion detection and prevention system. It supports high-performance multi-threaded packet processing, deep packet inspection, protocol identification, and file extraction from network traffic. Suricata's compatibility with the Emerging Threats and other community-driven rule sets enables detection of a wide range of known attack signatures, malware communications, exploit attempts, and suspicious network behavior. Its ability to operate efficiently in high-throughput environments makes it suitable for both small-scale laboratory setups and large-scale enterprise deployments.

The combination of Wazuh and Suricata represents a powerful open-source security stack capable of providing end-to-end visibility across both host and network layers. This integration enables correlation of endpoint events with network-based threats, significantly enhancing detection accuracy and improving the overall context available to security analysts. Such correlation is essential for identifying multi-stage attacks that involve reconnaissance, exploitation, lateral movement, and data exfiltration across different system layers.

One of the key advantages of the open-source security ecosystem is its flexibility and extensibility. Organizations can customize detection rules, develop custom plugins, integrate third-party tools, and modify system configurations according to their specific security requirements. This level of adaptability is often limited in proprietary security solutions, where customization options may be restricted or require additional licensing.

Another important strength of open-source security platforms is their strong community support. Continuous contributions from developers, researchers, and cybersecurity professionals help improve detection rules, fix vulnerabilities, and introduce new features. This collaborative development model ensures that open-source tools remain up to date with emerging threats and evolving attack techniques.

From a cost perspective, open-source security solutions significantly reduce the financial barriers associated with deploying enterprise-grade security infrastructure. Commercial SIEM and network security solutions often require substantial licensing fees, maintenance costs, and vendor dependencies. In contrast, open-source alternatives enable organizations to implement robust security monitoring capabilities while maintaining full control over infrastructure and budget allocation. This democratization of cybersecurity technology is particularly beneficial for small and medium-sized enterprises (SMEs), educational institutions, and research organizations.

Despite these advantages, open-source security platforms also present certain challenges. They often require higher levels of technical expertise for deployment, configuration, and maintenance compared to commercial solutions. Additionally, organizations must take responsibility for system updates, rule tuning, performance optimization, and integration management. Without proper configuration and ongoing maintenance, the effectiveness of these tools may be reduced.

In practical Security Operations Center (SOC) environments, open-source tools are increasingly being adopted either as standalone solutions or as part of hybrid architectures that combine open-source and commercial technologies. Their ability to provide transparency, customization, and cost efficiency makes them valuable components of modern cybersecurity strategies.

In the context of this project, the integration of Wazuh and Suricata demonstrates how open-source technologies can be leveraged to build a fully functional cybersecurity monitoring laboratory. The combined system provides centralized log management, endpoint monitoring, network intrusion detection, alert correlation, and visualization capabilities. This implementation highlights the practical viability of open-source security ecosystems in delivering enterprise-like functionality within controlled and cost-effective environments.

As cybersecurity threats continue to evolve, the open-source security ecosystem is expected to play an increasingly important role in global cyber defense strategies. Continued innovation, community collaboration, and integration with emerging technologies such as artificial intelligence and cloud-native architectures will further enhance the capabilities and adoption of these platforms in future security environments.

CHAPTER 3: TECHNOLOGIES USED

3.1 Wazuh

Wazuh is an open-source security platform that provides unified Security Information and Event Management (SIEM), Extended Detection and Response (XDR), and compliance monitoring capabilities. Originally forked from the OSSEC host-based intrusion detection system, Wazuh has evolved into a comprehensive security solution widely adopted by organizations, academic institutions, and research environments. The platform is released under the GNU General Public License (GPL) and the Apache License 2.0, ensuring transparency, flexibility, and community-driven development.

Wazuh is designed to provide centralized security monitoring by collecting, analyzing, and correlating security data from endpoints, network devices, applications, and cloud environments. It operates through a distributed architecture consisting of three primary components: Wazuh agents, Wazuh manager, and the Wazuh indexer and dashboard. The agents are installed on monitored endpoints and are responsible for collecting logs, monitoring system activity, and detecting suspicious behavior. The manager processes incoming data, applies detection rules, and generates alerts. The indexer stores and organizes security events, while the dashboard provides visualization and analysis capabilities for security analysts.

One of the key strengths of Wazuh is its ability to perform host-based intrusion detection. It continuously monitors endpoint activities such as file integrity changes, process execution, authentication attempts, registry modifications (on Windows systems), and system configuration changes. This enables the detection of unauthorized access, malware activity, privilege escalation attempts, and other indicators of compromise.

Wazuh also includes vulnerability detection capabilities by scanning installed software packages and comparing them against known vulnerability databases. This helps organizations identify outdated or vulnerable applications that may be exploited by attackers. Additionally, its compliance monitoring features support regulatory frameworks such as PCI-DSS, GDPR, HIPAA, and ISO 27001 by providing audit-ready reports and continuous compliance assessments.

Another important feature of Wazuh is its integration capability with external security tools. In this project, Wazuh is integrated with Suricata IDS to enhance network-level visibility. This integration enables correlation between host-based events and network-based threats, providing a more comprehensive understanding of attack patterns. Wazuh also supports integration with threat intelligence feeds, allowing it to detect indicators of compromise such as malicious IP addresses, domains, and file hashes.

The platform includes a powerful alerting and visualization system through the Wazuh Dashboard, which is built on OpenSearch. This interface allows security analysts to view real-time alerts, analyze historical

security data, investigate incidents, and generate security reports. Dashboards and visualizations play a critical role in Security Operations Center (SOC) workflows by improving situational awareness and enabling faster decision-making.

Wazuh also supports active response capabilities, which allow automated actions to be executed when specific security events are detected. These actions may include blocking malicious IP addresses, terminating suspicious processes, or executing custom scripts. This feature introduces a level of automation aligned with Security Orchestration, Automation, and Response (SOAR) principles.

In the context of this project, Wazuh serves as the core SIEM and XDR platform responsible for centralized log collection, event correlation, and alert generation. It provides endpoint monitoring through installed agents on Ubuntu systems and integrates with Suricata to incorporate network intrusion detection capabilities. The combination of these features enables comprehensive security monitoring across both host and network layers, forming the foundation of the proposed cybersecurity monitoring laboratory.

Overall, Wazuh plays a critical role in enabling real-time security visibility, threat detection, and incident analysis within the implemented system. Its open-source nature, extensibility, and wide feature set make it a suitable choice for academic research, laboratory environments, and small-to-medium enterprise deployments seeking cost-effective security solutions.

3.1.1 Core Functions

Wazuh provides the following key security functions in this project:

- **Log Collection and Analysis:** The Wazuh agent collects system logs, application logs, and security events from monitored endpoints and forwards them to the Wazuh Manager for analysis. It ensures centralized visibility across all connected systems, making it easier to correlate events and detect anomalies in real time.
- **Threat Detection:** The rule engine matches collected events against a library of over 3,000 built-in detection rules, generating alerts for suspicious activity. These rules are continuously updated to improve detection accuracy against emerging threats and attack patterns. It helps identify brute-force attempts, malware behavior, and unauthorized access quickly.
- **File Integrity Monitoring (FIM):** Wazuh monitors critical system files and directories for unauthorized modifications, providing early warning of potential compromise. It tracks changes such as file creation, deletion, and permission modifications. This helps detect insider threats and tampering with system configurations.
- **Vulnerability Assessment:** Integrated vulnerability detection identifies known CVEs affecting installed software packages. It scans system inventories regularly and maps them against global vulnerability databases. This enables organizations to prioritize patching based on risk severity.
- **Active Response** Wazuh can automatically execute response actions such as blocking IP addresses, killing processes, or triggering custom scripts when specific threats are detected. These automated responses reduce incident response time and help contain threats before they spread.

- **Centralized Security Dashboard:** Wazuh provides a unified dashboard to visualize alerts, compliance status, and system health. This improves security monitoring efficiency and simplifies decision-making for administrators.

3.1.2 Architecture Components

Component	Description
Wazuh Manager	Central processing component responsible for event analysis, rule matching, alert generation, and active response coordination.
Wazuh Indexer	Built on OpenSearch, provides scalable event storage, data indexing, and search capabilities for security event data.
Wazuh Dashboard	Web-based visualization interface built on OpenSearch Dashboards, providing alert monitoring, incident investigation, and compliance reporting.
Wazuh Agent	Lightweight software installed on monitored endpoints to collect and forward security events to the Wazuh Manager.

3.2 Suricata

Suricata is a high-performance, open-source network threat detection engine developed by the Open Information Security Foundation (OISF). It is widely used for intrusion detection (IDS), intrusion prevention (IPS), and network security monitoring (NSM) in modern cybersecurity environments. Suricata is designed to analyze network traffic in real time and detect malicious activities, policy violations, and suspicious behavior patterns across enterprise networks. It uses deep packet inspection (DPI) to examine packet contents beyond basic header information, enabling more accurate threat detection.

One of Suricata's key strengths is its multi-threaded architecture, which allows it to efficiently utilize modern multi-core processors. This enables the engine to process large volumes of network traffic at high speeds, including multi-gigabit network environments. It supports automatic protocol detection, allowing it to identify and analyze protocols such as HTTP, HTTPS, DNS, FTP, SMB, and TLS without requiring manual configuration.

Suricata uses rule-based detection to identify threats, relying on signature sets such as the Emerging Threats (ET) ruleset. These rules define patterns associated with known attacks, malware communications, port scans, denial-of-service attacks, and exploit attempts. When network traffic matches a rule, Suricata generates alerts that are logged for further analysis.

In addition to signature-based detection, Suricata also supports anomaly-based analysis and file extraction from network streams, which helps in identifying advanced threats. It can operate in both passive IDS mode and active IPS mode, where it can drop or block malicious traffic in real time.

Suricata generates logs in formats such as JSON, making it easy to integrate with SIEM platforms like Wazuh. In this project, Suricata is used to monitor network traffic, detect reconnaissance activities such as Nmap scans, and identify suspicious behaviors like ARP spoofing and unauthorized connections. Its integration with Wazuh enhances overall visibility by correlating network-level events with host-based security data.

Overall, Suricata plays a crucial role in providing network-level security monitoring, enabling early detection of threats and strengthening the overall cybersecurity posture of the system.

Key features utilized in this project include:

- **Deep Packet Inspection (DPI):** Suricata analyzes packet payloads beyond header information, enabling detection of application-layer attacks and protocol anomalies.
- **Signature-Based Detection:** Suricata uses the Emerging Threats (ET) rule set to detect thousands of known attack signatures, exploit attempts, and malicious network behaviors.
- **Network Traffic Analysis:** Detailed logging of network flows, HTTP transactions, DNS queries, and TLS certificates provides rich network context for threat investigations.
- **Eve JSON Logging:** Suricata outputs alerts and event logs in structured JSON format (EVE), facilitating integration with SIEM platforms such as Wazuh.

3.3 OpenSearch

OpenSearch is an open-source, distributed search and analytics engine derived from Elasticsearch. It is designed to handle large-scale data ingestion, storage, search, and analysis in real time. In the Wazuh architecture, OpenSearch functions as the Wazuh Indexer, responsible for storing and indexing security events generated from endpoints, network devices, and integrated security tools. It enables fast retrieval and correlation of security logs, which is essential for effective threat detection and incident investigation.

OpenSearch is built for horizontal scalability, allowing it to distribute data across multiple nodes in a cluster. This ensures high availability, fault tolerance, and efficient handling of large volumes of security data. It supports full-text search capabilities, enabling security analysts to quickly query logs and identify relevant events during investigations.

In addition to search functionality, OpenSearch provides powerful aggregation features that allow summarization and analysis of security events. This helps in identifying patterns such as repeated login failures, unusual network traffic spikes, or suspicious system behavior. It also supports real-time data ingestion and querying, ensuring that newly generated security events are immediately available for analysis.

OpenSearch Dashboards provides the visualization layer used in the Wazuh Dashboard. It offers interactive charts, graphs, and dashboards that help security analysts monitor system activity, investigate incidents, and generate reports. These visual tools improve situational awareness and support efficient decision-making in Security Operations Center (SOC) environments.

In this project, OpenSearch plays a critical role in enabling scalable storage, fast search performance, and effective visualization of security data within the Wazuh-based monitoring system.

3.4 Kali Linux

Kali Linux is a Debian-based Linux distribution specifically designed for penetration testing, ethical hacking, and security auditing. Developed and maintained by Offensive Security, it includes over 600 pre-installed security tools covering reconnaissance, vulnerability assessment, exploitation, and post-exploitation phases of a penetration test.

In this project, Kali Linux serves as the attacker machine used to simulate real-world cyber attacks. Tools utilized include:

- Nmap: The network mapper tool is used for host discovery, port scanning, and service enumeration to simulate reconnaissance attacks.
- arpspoof (from dsniff): Used to simulate ARP spoofing attacks for Man-in-the-Middle attack scenarios.
- netdiscover: Network address discovery tool used for host enumeration.

3.5 Ubuntu Server

Ubuntu Server 22.04 LTS serves as the operating system for the Wazuh SIEM platform and monitored endpoints. Ubuntu was selected for its stability, extensive package ecosystem, and strong community support. The Long Term Support (LTS) release provides five years of security updates, making it suitable for security-critical infrastructure. It offers strong compatibility with security tools such as Wazuh and Suricata, ensuring smooth integration and deployment. The lightweight nature of the server edition reduces system overhead, allowing better performance for security monitoring workloads. Ubuntu also provides robust command-line management capabilities, which are essential for configuring and maintaining server environments. Additionally, its widespread adoption in enterprise and cloud environments makes it a reliable choice for simulation of real-world security infrastructures.

CHAPTER 4: SYSTEM ARCHITECTURE

4.1 Architecture Overview

The cybersecurity monitoring laboratory implements a layered security architecture that mirrors real-world Security Operations Center (SOC) deployments. The architecture is designed to provide comprehensive visibility across both network and endpoint dimensions while maintaining clear separation between attacker, monitored, and monitoring components. This separation ensures realistic simulation of cyberattacks and accurate evaluation of detection capabilities within a controlled environment. The overall data flow follows a structured pipeline from event generation through detection, processing, storage, and visualization.

Component	Role in Laboratory Architecture
Attacker Machine	Kali Linux virtual machine is used as the attack platform to generate malicious traffic targeting the monitored Ubuntu endpoint. It performs reconnaissance and exploitation activities such as Nmap scanning, ARP spoofing, port scanning, and network enumeration to simulate real-world attack scenarios.
Suricata IDS	Suricata performs real-time network traffic analysis at the network layer. It inspects packets using deep packet inspection (DPI) and matches traffic patterns against the Emerging Threats rule set, generating alerts in EVE JSON format for detected malicious activities.
Wazuh Agent	The Wazuh agent installed on the monitored Ubuntu endpoint collects system logs, authentication events, file integrity monitoring data, and security-related events. It also forwards Suricata-generated alerts to the central Wazuh Manager for correlation and analysis.
Wazuh Manager	Acts as the central processing engine of the security monitoring system. It decodes incoming logs, applies correlation and detection rules, evaluates security events, and generates alerts. It also supports active response mechanisms for automated mitigation actions.
Wazuh Indexer	Security events and alerts are stored and indexed in OpenSearch through the Wazuh Indexer.
Wazuh Dashboard	Security analysts view and investigate alerts through the web-based dashboard with rich visualization capabilities.

4.2 Network Topology

The laboratory environment uses a virtual network topology implemented in VirtualBox. All virtual machines are connected through a host-only network adapter, providing isolated network communication between the attacker and victim machines without exposing the laboratory to external networks. This ensures that all generated traffic remains confined within the virtual environment, improving both safety and control during attack simulations.

The network is configured with a /24 subnet, providing sufficient address space for all laboratory components, including attacker, victim, and monitoring systems. The Wazuh SIEM server is assigned a static IP address to ensure consistent connectivity from the Wazuh agents and other integrated services. Static addressing also simplifies configuration of log forwarding, agent registration, and rule-based monitoring.

The Ubuntu server acting as the monitored endpoint is configured with a dynamic or static IP within the same subnet, allowing seamless communication with the Wazuh Manager. The Kali Linux attacker machine resides in the same virtual network segment to simulate real-world attack scenarios such as reconnaissance, scanning, and exploitation.

Network isolation is a key design principle in this topology, ensuring that malicious traffic generated during testing does not affect external systems or the host machine. This makes the environment safe for experimentation with intrusive attack techniques such as ARP spoofing and port scanning.

The host-only adapter also enables bidirectional communication between virtual machines while preventing direct internet exposure unless explicitly configured. This helps maintain a controlled and predictable testing environment, which is essential for accurate security analysis.

Additionally, all systems are configured to ensure time synchronization, which is critical for accurate log correlation across Suricata and Wazuh events. Proper time alignment improves forensic analysis and event timeline reconstruction during incident investigation.

The topology is designed to be scalable, allowing additional virtual machines such as web servers, database servers, or additional endpoints to be added for more complex attack simulations. This flexibility makes the laboratory suitable for advanced cybersecurity experiments and research scenarios.

Overall, the network topology provides a secure, isolated, and scalable environment that closely replicates enterprise network conditions while remaining suitable for academic and research purposes.

4.3 Component Interactions

The Wazuh Manager communicates with registered agents over an encrypted channel using the Wazuh proprietary protocol. Agents forward collected logs and events to the manager at configurable intervals

and in real time for high-priority events. The manager processes these events through its analysis engine, applying built-in and custom rules to generate security alerts.

Suricata IDS is configured to write EVE JSON logs to a shared directory accessible by the Wazuh agent. The Wazuh agent monitors this log file and forwards new Suricata alerts to the Wazuh Manager for processing and correlation with host-based events.

4.4 Data Flow

Security event data flows through the following stages: collection, transmission, analysis, storage, and visualization. Each stage is optimized to minimize latency while maintaining data integrity. Alert generation typically occurs within seconds of event detection, enabling near-real-time threat visibility. This rapid processing is critical for identifying and responding to active security threats before they escalate.

Data normalization is performed during the analysis stage to ensure consistency across logs from different sources and formats. This allows effective correlation between endpoint and network events within the Wazuh Manager. The system also prioritizes high-severity alerts to ensure faster response for critical incidents. Time synchronization across all components further improves accuracy in event correlation and forensic analysis. Overall, this pipeline ensures efficient and reliable security monitoring across the entire laboratory environment.

CHAPTER 5: ENVIRONMENT SETUP

5.1 Hardware Requirements

The laboratory environment is designed to run on commodity hardware, making it accessible for academic and small enterprise use. The minimum and recommended hardware specifications are as follows:

Component	Minimum	Recommended
RAM	8 GB	16 GB
Processor	Dual Core 2.4 GHz	Quad Core 3.0 GHz
Storage	50 GB SSD	100 GB SSD
Network	Single NIC	Dual NIC

5.2 Software Requirements

The following software components are required to build the laboratory environment:

Software	Version	Purpose
VirtualBox	7.0+	Virtualization platform for hosting VMs
Ubuntu Server	22.04 LTS	Wazuh SIEM server OS
Kali Linux	2024.x	Attacker machine OS
Wazuh	4.x	SIEM/XDR platform
Suricata	7.x	Network IDS engine
OpenSearch	2.x	Data indexing and search

5.3 Network Configuration

The virtual laboratory uses VirtualBox host-only networking to create an isolated environment. Both the Wazuh SIEM server and the Kali Linux attacker machine are connected to the same host-only network adapter, enabling network communication between the VMs without Internet exposure. This ensures a secure and controlled environment for simulating cyberattacks and monitoring security events.

Network configuration details:

- Host-only network subnet: 192.168.56.0/24
- Wazuh SIEM server: 192.168.56.10 (static)
- Kali Linux attacker: 192.168.56.20 (static)
- DNS: 8.8.8.8 (for external package installation)

The use of static IP addressing ensures consistent connectivity between the SIEM server and monitored systems. This is particularly important for maintaining stable agent-to-manager communication in Wazuh. The subnet provides sufficient address space for adding additional virtual machines such as additional endpoints or servers for extended testing scenarios.

Network isolation is enforced to prevent any external interference from the host machine or external networks. This improves the reliability of attack simulations and ensures that all generated traffic remains within the controlled laboratory environment. It also enhances safety when executing potentially disruptive attack techniques such as scanning or spoofing.

The configuration also supports bidirectional communication between attacker and victim machines, enabling realistic attack scenarios. All systems are configured within the same broadcast domain, which allows network-based attacks like ARP spoofing to be effectively demonstrated. Proper routing is intentionally avoided to maintain a simple flat network topology suitable for analysis.

DNS configuration using Google DNS (8.8.8.8) allows secure package updates and software installation when required. However, external connectivity is minimized during attack simulation phases to preserve isolation. This balance ensures both functionality and security within the laboratory setup.

Time synchronization and consistent network configuration across all virtual machines further improve log correlation accuracy within the Wazuh platform. Overall, the network setup provides a stable, scalable, and realistic environment for cybersecurity monitoring experiments.

5.4 Virtual Machine Configuration

Each virtual machine requires specific resource allocation to ensure stable operation. The Wazuh SIEM server requires the most resources due to the computational demands of the Wazuh Manager, Indexer, and Dashboard components running simultaneously. The Wazuh SIEM VM is configured with 8 GB RAM, 4 vCPUs, and a 50 GB virtual disk. The Kali Linux attacker VM requires 4 GB RAM, 2 vCPUs, and a 30 GB virtual disk. Both VMs use the VMDK disk format for compatibility and are configured with PAE/NX enabled for improved performance.

The allocated resources ensure smooth handling of log ingestion, indexing, and real-time alert processing without performance bottlenecks. Adequate memory allocation is particularly important for OpenSearch operations within the Wazuh Indexer. CPU allocation supports parallel processing of security events and IDS traffic analysis. Disk allocation provides sufficient storage for logs, alerts, and historical event data. Overall, this configuration balances performance and system stability within the virtual laboratory environment.

CHAPTER 6: WAZUH INSTALLATION AND CONFIGURATION

6.1 Installation Overview

Wazuh provides an automated installation script that deploys all three components — the Wazuh Manager, Indexer, and Dashboard — in a single-node configuration suitable for laboratory use. The installation process typically takes 15–30 minutes depending on system performance and internet connectivity.

The Wazuh documentation recommends a minimum of 4 GB RAM for the all-in-one installation, with 8 GB recommended for stable operation. The installation script configures all required services, generates SSL certificates for encrypted communication, and initializes the OpenSearch cluster.

This automated approach significantly simplifies deployment by reducing manual configuration errors and setup complexity. It ensures that all components are properly integrated and version-compatible during installation. The script also configures system services to start automatically on boot, ensuring persistence across system restarts. Additionally, default security settings such as firewall rules and authentication credentials are initialized to secure the environment. This makes the installation process suitable for rapid deployment in academic and laboratory environments

6.2 Wazuh Manager Configuration

Following installation, the Wazuh Manager requires configuration to support the desired monitoring capabilities. Key configuration files are located in `/var/ossec/etc/` and include `ossec.conf` for global settings and rules directories for detection logic.

Important configuration areas include:

- Email notification settings: Configure SMTP server details, sender/recipient addresses, and alert level thresholds for email-based alerting.
- Active response configuration: Define response scripts and trigger conditions for automated threat responses.
- Log analysis: Configure log file monitoring paths for Suricata EVE JSON output and custom application logs.
- Vulnerability detection: Enable and configure the vulnerability detector module for CVE scanning.

6.3 Wazuh Indexer Configuration

The Wazuh Indexer (OpenSearch) is pre-configured by the installation script with sensible defaults for single-node operation. Key settings include cluster name, node roles, JVM heap size, and network binding

address. These default configurations are optimized to ensure stable performance in laboratory and small-scale environments without requiring extensive manual tuning.

For production deployments, additional tuning is required for performance optimization, scalability, and high availability. Index lifecycle management (ILM) policies play a crucial role in controlling how security event data is retained, aged, and eventually deleted. These policies help manage storage efficiently by automatically transitioning indices through different stages such as hot, warm, and delete phases based on their age and usage patterns.

Default configurations typically retain indices for 90 days; however, this duration can be adjusted depending on organizational requirements, storage capacity, and regulatory compliance standards. In environments with high log volume, shorter retention periods may be necessary to optimize disk utilization, while compliance-driven environments may require longer retention for audit and forensic analysis purposes.

Proper index management also improves query performance by reducing the amount of active data that needs to be searched during analysis. Additionally, shard allocation and replication settings can be tuned in multi-node deployments to enhance fault tolerance and system resilience. Overall, these configurations ensure that the Wazuh Indexer remains efficient, scalable, and suitable for both educational and enterprise-grade security monitoring scenarios.

6.4 Dashboard Configuration

The Wazuh Dashboard provides the primary analyst interface for monitoring security events, investigating alerts, and generating compliance reports. After installation, the dashboard is accessible via HTTPS on port 443, ensuring secure communication between the user and the platform. The default administrator credentials must be changed immediately upon first login to prevent unauthorized access and strengthen system security.

The dashboard is built on OpenSearch Dashboards, providing interactive visualizations such as charts, graphs, timelines, and event summaries that help analysts quickly understand security trends and incidents. It enables real-time monitoring of alerts generated by both Wazuh Manager and integrated tools such as Suricata IDS.

Dashboard modules relevant to this project include: Security Events for alert monitoring, Threat Hunting for deep event analysis, Integrity Monitoring for reviewing File Integrity Monitoring (FIM) alerts, and Vulnerability Detection for identifying CVE-related findings on monitored endpoints.

In addition, the dashboard supports filtering and correlation of events based on time range, severity level, agent name, and rule category, allowing efficient investigation of incidents. Analysts can drill down into individual alerts to view detailed metadata, including source IP, affected host, rule description, and associated log data. This improves situational awareness and speeds up root cause analysis.

The visualization capabilities of the dashboard play a key role in Security Operations Center (SOC) workflows by enabling quick identification of attack patterns and system anomalies. It also supports

report generation for compliance audits and security assessments. Overall, the Wazuh Dashboard serves as a centralized interface that enhances visibility, usability, and effectiveness of the entire security monitoring system.

6.5 Agent Enrollment

Wazuh agents are installed on monitored endpoints and registered with the Wazuh Manager using a secure enrollment process. The agent receives a unique ID and authentication key from the Manager, which is used to authenticate subsequent communications and ensure secure data exchange between endpoints and the central server.

Agent installation on Ubuntu involves downloading the Wazuh agent package from the official repository, configuring the Manager IP address, and starting the agent service. Once installed, the agent establishes an encrypted connection with the Wazuh Manager to ensure confidentiality and integrity of transmitted security data.

After successful registration, the agent automatically begins collecting and forwarding system events within seconds. These events include authentication logs, system activity logs, application logs, and security-related events generated by the endpoint operating system.

The agent also supports continuous monitoring features such as File Integrity Monitoring (FIM), log analysis, and vulnerability detection, depending on the configuration applied. It runs as a lightweight background service to minimize system overhead while maintaining real-time monitoring capabilities.

In addition, the agent periodically communicates with the Wazuh Manager to send heartbeat signals, ensuring that the system can detect offline or disconnected endpoints. This improves reliability and helps maintain accurate visibility of all monitored systems within the security infrastructure.

CHAPTER 7: SURICATA IDS INTEGRATION

7.1 Suricata Installation

Suricata is installed on the same Ubuntu server as the Wazuh components, enabling it to monitor network traffic at the server level. Installation is performed using the Open Information Security Foundation (OISF) Personal Package Archive (PPA) repository, which provides the latest stable releases for Ubuntu-based systems. This approach ensures that the installation remains up to date with the latest security patches, protocol support, and performance improvements.

Post-installation, Suricata requires configuration to specify the monitoring network interface, home network address range, and EVE JSON output settings. The `suricata.yaml` configuration file controls all aspects of Suricata operation, including detection engine settings, logging formats, rule paths, and protocol parsing options.

Proper configuration of the network interface is essential to ensure that Suricata captures all relevant traffic flowing through the system. The home network variable defines the trusted internal network range, allowing Suricata to distinguish between internal and external traffic patterns. This distinction is important for reducing false positives and improving detection accuracy.

The EVE JSON output format is enabled to generate structured logs that can be easily consumed by external tools such as Wazuh. These logs include detailed information about alerts, HTTP requests, DNS queries, TLS handshakes, and file extraction events, providing deep visibility into network activity.

Suricata rules are managed through rule files such as the Emerging Threats (ET) rule set, which contains signatures for known attack patterns, malware communication, and exploit attempts. Regular updates to these rules are necessary to maintain detection effectiveness against evolving threats.

In this project, Suricata operates in Intrusion Detection System (IDS) mode, passively analyzing network traffic without actively blocking packets. This configuration ensures safe monitoring within the laboratory environment while still providing accurate detection of malicious activities.

Overall, the Suricata installation and configuration process establishes the foundation for network-level threat detection and plays a crucial role in integrating network security monitoring with the Wazuh-based SIEM architecture.

7.2 Rule Management

Suricata uses `Suricata-Update`, an official rule management utility, to download, install, and maintain detection rule sets. This tool simplifies the process of keeping Suricata's detection capabilities up to date

by automating rule retrieval and configuration. It ensures that the IDS remains effective against newly emerging threats without requiring manual rule management.

The Emerging Threats Open (ET Open) rule set is used in this project, providing a comprehensive collection of more than 40,000 detection rules. These rules cover a wide range of attack categories, including network reconnaissance, vulnerability exploitation, malware command-and-control (C2) communication, denial-of-service attacks, brute-force attempts, and data exfiltration activities. This extensive coverage enables Suricata to detect both common and advanced attack techniques in real time.

Rules are updated regularly to maintain detection accuracy and ensure coverage of newly discovered vulnerabilities and attack patterns. The update process retrieves the latest rule packages from trusted repositories, validates them, and integrates them into the existing Suricata configuration.

During this process, Suricata-Update merges externally downloaded rules with any locally defined custom rules created for the laboratory environment. This allows administrators to extend detection capabilities by adding project-specific signatures tailored to simulated attack scenarios.

Once updated, the rule set is reloaded dynamically without requiring a full system restart, ensuring continuous network monitoring with minimal or no service interruption. This capability is particularly important in real-time security environments where downtime could result in missed malicious activity.

In this project, rule management plays a critical role in ensuring that Suricata remains responsive to evolving attack simulations performed using tools such as Kali Linux. The combination of community-driven rule sets and custom-defined rules enhances detection accuracy and provides a robust foundation for network-based intrusion detection within the cybersecurity monitoring laboratory.

7.3 EVE JSON Log Generation

Suricata's EVE JSON (Extensible Event Format) logging provides structured, machine-readable output for all detected network events. This format is designed to enable easy integration with external security tools and SIEM platforms by providing consistent and detailed event metadata.

EVE JSON records include comprehensive information for each event type. Alert records contain details such as the triggering rule signature, source and destination IP addresses, source and destination ports, protocol type, timestamp, and severity level. These fields allow security analysts to quickly understand the nature and context of detected threats.

Flow records provide information about network connections, including session duration, total bytes transferred, packet counts, and connection states. This helps in analyzing traffic patterns and identifying abnormal communication behavior between hosts.

DNS records capture query and response details, including queried domain names, resolved IP addresses, and response types. This is particularly useful for detecting malicious domains, command-and-control communication, and DNS-based tunneling attempts.

In addition to these, Suricata can also generate HTTP, TLS, and file extraction logs depending on configuration, providing deeper visibility into application-layer traffic. These structured logs enhance the ability to perform forensic analysis and threat hunting activities.

The EVE JSON output is written to `/var/log/suricata/eve.json`, which serves as the primary log source for network-based event data in this project. The Wazuh agent continuously monitors this file in real time and forwards newly generated entries to the Wazuh Manager for processing, correlation, and alert generation.

This integration enables seamless communication between Suricata and Wazuh, allowing network-level security events to be combined with host-based logs. As a result, security analysts gain a unified and enriched view of attack activities across both network and endpoint layers within the cybersecurity monitoring laboratory.

7.4 Wazuh Integration

Integration between Suricata and Wazuh is achieved by configuring the Wazuh agent to monitor the Suricata EVE JSON log file in real time. The agent continuously reads newly generated entries from `/var/log/suricata/eve.json` and forwards them securely to the Wazuh Manager for processing and analysis. This mechanism ensures that network-based security events are seamlessly ingested into the centralized SIEM environment.

The Wazuh Manager includes built-in decoders and rules specifically designed to interpret Suricata-generated events. These rules parse the JSON structure, extract relevant fields, and map Suricata alert severity levels to corresponding Wazuh alert levels. This standardization allows consistent classification and prioritization of security events across both network and host-based sources.

This integration enables advanced event correlation between Suricata network alerts and Wazuh endpoint alerts, providing a unified and enriched view of security incidents. By combining data from multiple layers, the system can reconstruct the full attack chain rather than viewing isolated events independently.

For example, a network reconnaissance activity such as an Nmap scan detected by Suricata can be correlated with subsequent suspicious behavior on the target system, such as unusual process execution, authentication attempts, or file system modifications detected by the Wazuh agent. This correlation helps identify multi-stage attack scenarios and improves detection accuracy.

The integration also reduces alert fatigue by consolidating related events into meaningful security incidents, rather than generating multiple independent alerts. This improves situational awareness for security analysts and enhances incident response efficiency.

Overall, the Wazuh–Suricata integration forms a critical component of the cybersecurity monitoring laboratory by bridging network-level and host-level visibility, enabling comprehensive threat detection and analysis within a unified security framework.

CHAPTER 8: ATTACK SIMULATION

8.1 Nmap Scan Detection

Nmap (Network Mapper) is one of the most widely used tools in both legitimate network administration and malicious reconnaissance. Attackers use Nmap to discover live hosts, identify open ports, enumerate running services, and detect operating system versions — all critical information for planning subsequent attack phases. This type of reconnaissance is often considered the first stage in the cyber kill chain, as it helps attackers map the target environment before launching exploitation attempts.

In addition to basic scanning, Nmap supports advanced techniques such as aggressive scanning, script-based enumeration (NSE scripts), and firewall evasion methods, making it a powerful tool for both penetration testing and malicious reconnaissance activities. In real-world scenarios, such scanning activities are often stealthy and distributed to avoid detection by security systems.

8.1.1 Attack Scenario

The Nmap scan simulation involves launching a comprehensive scan from the Kali Linux attacker machine targeting the Ubuntu SIEM server. The scan includes SYN stealth scan (-sS), service version detection (-sV), and OS detection (-O), generating significant network traffic that Suricata is configured to detect.

The scan is executed in multiple phases to simulate realistic attacker behavior, starting with host discovery followed by port enumeration and service fingerprinting. This ensures that the generated traffic closely resembles real-world reconnaissance patterns rather than isolated test packets. The attacker machine acts as an external adversary attempting to gather intelligence about the target system's exposed services and configurations.

This scenario is designed to evaluate the effectiveness of network-based intrusion detection in identifying both basic and advanced scanning techniques under controlled laboratory conditions.

8.1.2 Detection Process

The detection process for Nmap scanning follows these sequential steps:

- The Nmap scan is launched from the Kali Linux machine, generating probe packets directed at the target Ubuntu server.
- Suricata detects the characteristic patterns of Nmap probing — specifically the rapid sequential SYN packets to multiple ports — and generates ET SCAN rule alerts.
- The Suricata alerts are written to the EVE JSON log file and forwarded by the Wazuh agent to the Wazuh Manager.

- The Wazuh Manager processes the Suricata alerts through its rule engine, applying additional correlation rules and generating high-severity security alerts.
- Alerts appear on the Wazuh Dashboard in near real time, providing the analyst with full visibility of the reconnaissance activity including attacker IP, targeted ports, and scan technique.

8.2 Port Scan Detection

Port scanning is a precursor to most network attacks, allowing adversaries to identify potential entry points by discovering open services. Beyond Nmap, attackers may use tools such as Masscan, Zmap, or custom scripts to perform high-speed port enumeration. These tools are often used in large-scale automated attack campaigns due to their speed and efficiency.

Suricata detects port scanning through multiple rule categories: ET SCAN rules detect specific tool signatures, threshold rules identify excessive connection attempts to multiple ports from a single source, and stream rules detect incomplete TCP handshakes characteristic of SYN scanning. The combination of these detection methods provides robust coverage across different scanning techniques and tools.

In addition to signature-based detection, behavioral analysis plays an important role in identifying stealth scans that attempt to evade traditional rules. For example, slow-rate scans distributed over time can still be detected using anomaly thresholds and connection frequency analysis.

Port scan detection is critical because it often indicates the initial stage of a larger attack campaign. Early detection enables security teams to proactively block malicious sources before exploitation attempts begin. In this project, port scan alerts are correlated with endpoint activity logs to determine whether scanning activity leads to further compromise attempts.

8.3 Alert Analysis and Investigation

Once alerts are generated, the Wazuh Dashboard provides powerful tools for investigation and forensic analysis. The Security Events module displays alerts in real time with filtering and sorting capabilities. Each alert provides detailed information including the timestamp, rule ID, rule description, alert severity, source and destination information, and the raw event data.

Analysts can use filtering options such as IP address, rule category, severity level, or time range to narrow down relevant events during investigations. This helps in quickly identifying attack patterns and understanding the sequence of malicious activities.

The Threat Hunting module enables more advanced investigation by querying the indexed event data with custom filters. Analysts can build timelines of attacker activity, pivot from network events to host events, and export findings for reporting. This capability is essential for reconstructing multi-stage attacks and identifying root causes of security incidents.

In addition, correlation features within Wazuh allow related alerts to be grouped into a single incident view, reducing alert fatigue and improving situational awareness. Analysts can also visualize attack trends over time using dashboards, which helps in identifying recurring threats or persistent attackers.

Overall, the alert analysis and investigation phase plays a crucial role in transforming raw security data into actionable intelligence, enabling effective decision-making and incident response.

CHAPTER 9: ARP SPOOFING ATTACK SIMULATION

9.1 Introduction to ARP Spoofing

Address Resolution Protocol (ARP) is a fundamental Layer 2 networking protocol used to map IP addresses to corresponding Media Access Control (MAC) addresses within a local network segment. When a device needs to communicate with another device on the same network, it relies on ARP to determine the destination MAC address associated with a known IP address. However, ARP was designed without authentication or verification mechanisms, making it inherently vulnerable to malicious manipulation. Since devices automatically trust and accept ARP responses, attackers can exploit this weakness by sending forged ARP messages to other hosts on the network.

ARP spoofing, also known as ARP poisoning or ARP cache poisoning, is a technique in which an attacker transmits falsified ARP reply packets to associate their own MAC address with the IP address of a legitimate device, typically the default gateway. As a result, network traffic intended for the legitimate device is redirected through the attacker's system. This allows the attacker to position themselves between communicating devices and perform a Man-in-the-Middle (MitM) attack. Once the attack is successful, the adversary can intercept, monitor, modify, or block network communications without the knowledge of the victims.

The effectiveness of ARP spoofing stems from the trust-based nature of the ARP protocol and its widespread use in IPv4 networks. Through this attack, sensitive information such as login credentials, session cookies, personal data, and confidential communications can be captured, particularly when transmitted through unsecured protocols. In addition to traffic interception, ARP spoofing can also be used to facilitate session hijacking, denial-of-service attacks, and traffic redirection. Attackers often maintain the attack by continuously sending forged ARP messages to ensure that poisoned entries remain in the victims' ARP caches.

Because ARP spoofing operates within the local network environment, it is especially dangerous in shared networks where multiple devices communicate within the same broadcast domain. Although encrypted protocols such as HTTPS, SSH, and VPNs can help protect the confidentiality of transmitted data, they do not completely eliminate the risks associated with traffic manipulation. Consequently, organizations employ various countermeasures, including static ARP entries, Dynamic ARP Inspection (DAI), network segmentation, port security mechanisms, and intrusion detection systems to mitigate the threat. Understanding ARP spoofing is essential for evaluating network security weaknesses and implementing effective defenses against Layer 2 attacks.

9.2 Attack Workflow

In this laboratory environment, the ARP spoofing attack is performed in a controlled setting to demonstrate how an attacker can position themselves between a victim machine and the network gateway. The attack begins by configuring the attacker system, running Kali Linux, to enable IP forwarding. This

step is essential because it allows the attacker's machine to forward intercepted packets to their intended destinations, ensuring that network communication remains uninterrupted and reducing the likelihood of detection by the victim.

Once IP forwarding is enabled, the attacker initiates the ARP spoofing process by sending forged ARP reply messages to the victim machine. These malicious ARP responses falsely claim that the IP address of the default gateway is associated with the attacker's MAC address. As a result, the victim updates its ARP cache and begins sending traffic intended for the gateway directly to the attacker's system. Simultaneously, the attacker sends additional forged ARP replies to the gateway, falsely indicating that the victim's IP address corresponds to the attacker's MAC address. This causes the gateway to also update its ARP cache with the incorrect mapping.

Through these forged ARP associations, both the victim and the gateway mistakenly direct their traffic to the attacker's machine. Consequently, all communication exchanged between the two devices passes through the attacker, effectively establishing a Man-in-the-Middle (MitM) position. In this state, the attacker can observe network traffic, capture sensitive information, analyze communication patterns, or manipulate packets before forwarding them to their intended destination. To maintain the attack, forged ARP replies are continuously transmitted, ensuring that the poisoned ARP cache entries remain active and are not replaced by legitimate mappings.

This workflow demonstrates how the absence of authentication mechanisms within the ARP protocol can be exploited to redirect network traffic and compromise the confidentiality and integrity of communications within a local network.

9.3 Detection Strategy

Detection of ARP spoofing in this environment leverages multiple monitoring layers. Network anomaly detection is implemented using Suricata, which identifies unusual ARP traffic patterns such as gratuitous ARP replies, excessive broadcast activity, and MAC address inconsistencies that may indicate spoofing attempts. Log correlation through the Wazuh Manager enhances visibility by combining ARP-related network alerts with endpoint and system logs, allowing for a more complete understanding of potential man-in-the-middle activity.

Traffic analysis further supports detection by examining network flow records and comparing expected communication paths with observed traffic routes to identify any abnormal redirection. In addition to these methods, continuous ARP table monitoring on endpoints helps detect unexpected changes in IP-to-MAC mappings, particularly for critical devices such as the default gateway.

MAC address validation against a known baseline of authorized devices provides another layer of assurance by highlighting unauthorized or unexpected hardware identities on the network. Packet

inspection techniques are also used to analyze ARP packets for irregularities in frequency, duplication of IP assignments, or inconsistencies in sender information.

Behavioral analysis contributes by establishing a normal network baseline and identifying deviations such as unusual latency, abnormal packet forwarding behavior, or unexpected communication patterns that often accompany ARP spoofing attacks. Gateway integrity checks are periodically performed to ensure that the default gateway's IP-to-MAC binding remains consistent and has not been impersonated.

Finally, alert aggregation and correlation across Suricata, Wazuh, network devices, and endpoint agents improves detection accuracy by reducing false positives and strengthening confidence in identifying genuine spoofing events. Historical analysis of ARP activity further supports forensic investigation by revealing recurring patterns or persistent threats over time, making the overall detection strategy more robust and reliable.

9.4 Security Risks and Impact

ARP spoofing poses significant security risks to organizations that do not implement appropriate countermeasures. Successful ARP spoofing attacks can enable credential theft, where unencrypted credentials transmitted over protocols such as HTTP, FTP, Telnet, and IMAP are exposed to the attacker during interception. This allows attackers to gain unauthorized access to internal systems and services.

Session hijacking is another major impact, where web session cookies captured through ARP spoofing can be reused to impersonate authenticated users without requiring the original login credentials, effectively bypassing authentication mechanisms.

Traffic manipulation is also a critical risk, as attackers positioned in the communication path can alter unencrypted HTTP traffic in transit. This may include injecting malicious scripts, modifying data being exchanged, or redirecting users to phishing websites designed to steal sensitive information.

Denial of service conditions can occur when the attacker intentionally drops or fails to forward intercepted packets, resulting in disrupted connectivity between victims and legitimate network resources. This can lead to service downtime and degraded network performance.

In addition to these direct impacts, ARP spoofing can also facilitate lateral movement within the network by allowing attackers to observe internal communication patterns, identify valuable targets, and escalate attacks to other systems. It can further undermine trust in network infrastructure, as users and systems may unknowingly communicate through a compromised path, making detection and response more difficult if monitoring controls are insufficient.

9.5 Mitigation Strategies

Several technical controls can be implemented to mitigate ARP spoofing attacks and reduce their impact on network security. One of the most effective methods is the use of static ARP entries for critical

network resources, which ensures that key IP-to-MAC mappings cannot be dynamically altered, thereby preventing ARP cache poisoning for essential systems such as gateways and servers.

Dynamic ARP Inspection (DAI) is another strong preventive control, typically enforced at the switch level. DAI validates ARP packets against a trusted DHCP snooping binding table and drops any forged or suspicious ARP replies, effectively blocking unauthorized MAC-to-IP associations before they propagate through the network.

Network segmentation through VLANs further reduces risk by isolating network traffic into smaller, controlled broadcast domains. This limits the scope of ARP spoofing attacks, ensuring that even if an attack succeeds within one segment, it does not easily extend across the entire network infrastructure.

Encryption plays a critical role in reducing the effectiveness of intercepted traffic. The universal adoption of TLS/SSL for application-layer communication ensures that even if ARP spoofing enables packet interception, the attacker cannot easily decrypt or manipulate the protected data without valid session keys.

Intrusion detection system (IDS) monitoring, such as continuous Suricata-based analysis, provides real-time visibility into suspicious ARP activity. This allows for early detection and response to ongoing spoofing attempts, minimizing potential damage and enabling rapid incident containment.

CHAPTER 10: XDR CAPABILITIES

10.1 Extended Detection Overview

Extended Detection and Response (XDR) represents the evolution of EDR (Endpoint Detection and Response) toward a unified security detection and response platform that integrates data from multiple security layers. Wazuh's XDR capabilities are realized through the combination of its endpoint agent framework with Suricata network monitoring, creating a platform that correlates threats across host and network dimensions. This integrated approach improves detection accuracy by combining telemetry from different sources into a single analytical framework.

In addition, XDR enables automated correlation of seemingly unrelated events, allowing security teams to detect multi-stage attacks that would otherwise remain hidden. It also enhances incident response by providing a centralized view of alerts across endpoints, networks, and applications, reducing the time required to identify and respond to threats.

Furthermore, the XDR model improves scalability in security operations by reducing the dependency on isolated security tools. Instead of analyzing endpoint and network data separately, XDR consolidates them into a unified pipeline that supports faster detection, investigation, and remediation.

10.2 Endpoint Monitoring

The Wazuh agent provides comprehensive endpoint monitoring capabilities that feed the XDR detection engine. Agent-based monitoring includes operating system log collection (syslog, auth.log, dpkg.log), file integrity monitoring of critical system paths, rootkit and malware detection through system call monitoring, and process execution monitoring for suspicious command patterns.

In addition, endpoint monitoring tracks user activity and authentication events to identify unauthorized access attempts or privilege escalation behavior. It also provides real-time visibility into system configuration changes, which helps in detecting unauthorized modifications that may indicate compromise. Endpoint telemetry is further enhanced through periodic health checks of installed software, allowing detection of outdated or vulnerable packages. This strengthens overall endpoint security posture by ensuring compliance with system hardening policies. This endpoint telemetry is enriched with contextual information including the agent hostname, operating system version, and IP addresses before being forwarded to the Wazuh Manager for analysis.

10.3 Network Monitoring

Suricata integration extends Wazuh's visibility to the network layer, providing alert data, flow records, and protocol logs that complement the endpoint data. Network monitoring detects threats that may not

generate host-level artifacts, such as reconnaissance scanning, network-layer denial of service attacks, and exploitation attempts against network services.

It also enables detection of anomalous traffic patterns such as unusual port usage, abnormal packet volumes, or repeated connection attempts, which may indicate early-stage intrusion attempts. Protocol-level inspection further helps identify malformed or suspicious packets that could be used in exploitation attempts.

Additionally, network monitoring supports detection of lateral movement within internal networks by identifying unusual communication between hosts that do not normally interact. This is particularly useful for identifying compromised systems spreading within an environment.

The combination of endpoint and network monitoring closes the detection blind spots that exist when either layer is monitored in isolation.

10.4 Centralized Detection and Correlation

The Wazuh Manager's rule engine enables cross-layer correlation of security events. Multi-source alerts combine network and endpoint events to identify attack campaigns with higher confidence than single-source detections would permit.

It also reduces false positives by validating suspicious activity across multiple data sources before generating high-severity alerts. This improves the accuracy and reliability of threat detection, especially in complex environments with high event volumes.

For example, an Nmap scan detected at the network level combined with a subsequent SSH brute force attempt detected at the endpoint level can be correlated as a coordinated attack sequence. Wazuh's threat intelligence integration enables enrichment of detected events with reputation data from external threat intelligence feeds, providing additional context for analyst investigations.

In addition, correlation rules can be customized to detect organization-specific attack patterns, making the system adaptable to different security requirements and threat models. This flexibility enhances its effectiveness in diverse operational environments

10.5 Threat Visibility

The Wazuh Dashboard serves as the single pane of glass for XDR threat visibility. The Overview module provides a high-level summary of security status across all monitored systems, including alert volumes, top alert categories, geographic threat origin maps, and agent health status. This centralized view enables analysts to quickly assess the security posture of the monitored environment and prioritize investigation efforts.

It also provides time-based analytics that allow security teams to identify trends in attack frequency and severity over time. This helps in understanding whether the threat landscape is improving or deteriorating.

Additionally, customizable dashboards allow different stakeholders, such as security analysts and administrators, to focus on metrics relevant to their roles. Drill-down capabilities further enable detailed investigation of individual alerts and events.

The integration of visualization, correlation, and reporting within a single interface significantly improves situational awareness and accelerates incident response workflows.

CHAPTER 11: EMAIL-BASED SOAR IMPLEMENTATION

11.1 Objective

Automated notification is a foundational SOAR capability that ensures security analysts receive timely alerts for critical security events, even when they are not actively monitoring the Wazuh Dashboard. The email notification system implemented in this project delivers alert details directly to analyst inboxes, enabling rapid awareness and response initiation. This improves incident response time by ensuring that high-severity alerts are communicated immediately across multiple channels.

In addition, the email-based alerting mechanism supports continuous monitoring by extending security visibility beyond the SOC dashboard environment. It ensures that incidents are not missed during off-hours or periods of reduced analyst activity, thereby improving overall operational resilience.

Furthermore, the system enhances prioritization of security events by allowing alerts to be categorized based on severity levels, ensuring that critical threats such as intrusion attempts, privilege escalation, or network attacks are escalated appropriately. It also serves as a bridge toward full SOAR automation by laying the groundwork for future integration with automated response workflows such as ticketing systems or script-based remediation actions.

11.2 Alert Notification Workflow

The email notification workflow follows a structured pipeline from event detection to analyst notification. The process begins when a security event occurs on a monitored endpoint or is detected by Suricata at the network level, such as suspicious process execution, unauthorized access attempts, or abnormal traffic patterns.

Once detected, the Wazuh Manager processes the event through its rule engine, where it is analyzed, normalized, and correlated with existing logs. If the event severity exceeds the configured threshold, a security alert is generated and categorized based on its risk level and relevance.

Following alert generation, Wazuh's email notification module evaluates the alert against predefined filtering rules. These rules determine whether the event requires immediate notification, ensuring that only high-priority or policy-defined alerts are forwarded to analysts to avoid unnecessary noise.

If the alert meets the criteria, an email notification is triggered and delivered through an SMTP server to the configured security analyst email address. The email typically includes key details such as the type of attack, affected host, timestamp, severity level, and suggested initial response actions for faster triage.

Finally, the analyst receives the notification and initiates the appropriate incident response procedure, which may include further investigation in the Wazuh Dashboard, endpoint inspection, or escalation to higher-level SOC workflows depending on the severity and impact of the event.

Stage	Description
Security Event	A security-relevant event occurs on a monitored endpoint or is detected by Suricata on the network.
Wazuh Alert	The Wazuh Manager processes the event through its rule engine and generates a security alert above the configured notification threshold.
Email Trigger	Wazuh's email notification module evaluates the alert against configured filters and determines whether an email notification should be sent.
SMTP Delivery	The notification is sent via SMTP to the configured analyst email address, containing alert details and recommended actions.
Analyst Response	The receiving analyst reviews the alert and initiates the appropriate incident response procedure.

11.3 Configuration

Wazuh's email notification system is configured in the `ossec.conf` file under the `<global>` section. Key parameters include the SMTP server address and port, sender email address, recipient email addresses, minimum alert level for notifications (typically level 10 for critical alerts), and email format settings. These configurations ensure that the email alerting mechanism is properly integrated with the organization's mail infrastructure and operates reliably for security event dissemination.

In addition, authentication settings such as SMTP username and password (if required by the mail server) can be defined to secure outbound email communication. This ensures that only authorized systems are able to send alert notifications, reducing the risk of misuse or spoofing of security emails.

Multiple email recipients can be configured, enabling simultaneous notification of multiple analysts or on-call personnel. This supports team-based incident response and ensures that alerts are not dependent on a single point of contact. Distribution lists can also be used to simplify management of large SOC teams.

Alert-level filtering ensures that only high-severity alerts generate email notifications, preventing analyst inbox flooding from low-priority events. This filtering mechanism improves operational efficiency by reducing alert fatigue and allowing analysts to focus on meaningful security incidents. Additionally, time-based or rule-based filtering can be applied to further refine which alerts are escalated, enhancing the precision of the notification system.

11.4 Alert Email Content

Each email notification includes structured alert information to enable rapid triage and efficient incident response by security analysts. The content is designed to provide both high-level context and detailed

forensic data so that the nature and impact of the event can be quickly assessed without requiring immediate access to the Wazuh Dashboard.

The email contains a timestamp indicating the exact date and time when the alert was generated, allowing analysts to correlate the event with other logs and activities across the environment. It also includes the Wazuh Rule ID, which identifies the specific detection rule that triggered the alert and helps in understanding the detection logic behind the event.

The severity level is provided using the Wazuh alert scale (1–15), enabling quick prioritization of incidents based on their potential impact. A higher severity level typically indicates more critical threats requiring immediate attention. In addition, a human-readable event description summarizes the detected threat or suspicious activity in a clear and understandable format.

Source information is also included, detailing the source IP address, hostname, and agent identifier associated with the event. This information is essential for identifying the affected system and initiating containment or investigation procedures. Finally, the raw log entry is attached within the email, providing complete unprocessed event data that supports deeper forensic analysis and validation of the alert.

- **Timestamp:** The exact date and time when the alert was generated.
- **Wazuh Rule ID:** The unique identifier of the detection rule that triggered the alert.
- **Severity Level:** The Wazuh alert level (1-15) indicating the severity of the detected event.
- **Event Description:** A human-readable description of the detected threat or suspicious activity.
- **Source Information:** The source IP address, hostname, and agent identifier associated with the event.
- **Raw Log:** The original log entry that triggered the alert, providing full context for investigation.

11.5 Benefits of Automated Notification

Email-based automated notification provides significant operational benefits for security teams by improving responsiveness, visibility, and overall efficiency in handling security incidents. It ensures that critical alerts are communicated immediately, reducing delays between detection and analyst awareness and thereby improving Mean Time to Respond (MTTR).

Another key advantage is 24/7 coverage, as email notifications ensure that analysts and on-call personnel receive alerts even outside regular working hours. This enables faster response to after-hours incidents and supports continuous security monitoring in environments where round-the-clock dashboard supervision is not always feasible.

Automated email alerts also contribute to documentation and auditability by creating a persistent record of security events within mail systems. These records can be referenced during incident investigations, compliance audits, and post-incident reviews, improving transparency and accountability in security operations.

In addition, this approach reduces dependency on continuous dashboard monitoring. Analysts are not required to constantly observe the Wazuh Dashboard, allowing them to allocate more time to deeper investigation, threat hunting, and remediation activities rather than passive monitoring of alerts.

Furthermore, email-based notifications improve collaboration within Security Operations Center (SOC) teams by enabling alerts to be shared instantly across multiple stakeholders. This ensures that incidents can be escalated quickly and handled by the appropriate personnel based on severity and expertise.

It also enhances prioritization of security incidents by enabling analysts to quickly filter and act on high-severity alerts directly from their inbox, reducing cognitive load and helping maintain focus on critical threats during high-volume attack scenarios.

CHAPTER 12: RESULTS AND ANALYSIS

12.1 Detection Results

The cybersecurity monitoring laboratory successfully demonstrated detection capabilities across all simulated attack scenarios. The integrated SIEM and IDS environment effectively identified both network-based and host-based attacks with low detection latency and appropriate severity classification, validating the effectiveness of the proposed architecture.

The following detection outcomes were achieved during testing:

Attack Type	Detection Tool	Alert Level	Detection Latency	Result
Nmap SYN Scan	Suricata + Wazuh	High (L12)	< 5 seconds	Detected
Nmap OS Detection	Suricata + Wazuh	Medium (L8)	< 5 seconds	Detected
Port Scan (TCP)	Suricata	High (L10)	< 3 seconds	Detected
ARP Spoofing	Suricata	High (L11)	< 10 seconds	Detected
SSH Brute Force	Wazuh Agent	Critical (L14)	< 5 seconds	Detected
File Integrity	Wazuh FIM	High (L11)	< 60 seconds	Detected

In addition to successful detection, the system demonstrated consistent correlation between network and endpoint alerts, particularly for reconnaissance and brute-force activities, where multiple detection layers reinforced alert confidence. This multi-layer validation helped reduce false positives and ensured accurate classification of threats.

Furthermore, the low detection latency across most attack types indicates that the system is capable of near real-time monitoring, which is essential for effective incident response in Security Operations Center (SOC) environments. The slightly higher latency observed in file integrity monitoring is expected due to scheduled scanning intervals and baseline comparison processing.

Overall, the results confirm that the implemented environment provides strong visibility across both network and host layers, enabling timely and reliable detection of common attack vectors in modern enterprise networks.

12.2 Dashboard Visualization

The Wazuh Dashboard provided comprehensive visualization of all detected security events throughout the testing period, enabling effective monitoring, analysis, and interpretation of security activity. The Security Events module displayed a real-time feed of alerts with advanced filtering, sorting, and drill-

down capabilities, allowing analysts to quickly isolate specific events based on severity, rule ID, attack type, or affected host.

The Overview dashboard provided a high-level summary of system security posture, including total alert volumes, distribution of top attack categories, and overall agent health status. This allowed for rapid assessment of the environment and helped identify periods of increased malicious activity or abnormal network behavior.

Alert visualizations included timeline charts that illustrated alert frequency trends over time, making it easier to correlate spikes in activity with simulated attack scenarios. Pie charts categorized alerts based on rule groups and severity levels, providing a clear breakdown of the types and seriousness of detected threats.

In addition, geographic mapping visualizations displayed the source IP locations of detected attacks, offering contextual insight into potential external threat origins. Agent status indicators further showed monitoring coverage across all enrolled endpoints, ensuring that all systems were actively reporting and contributing to the detection pipeline.

Overall, these visualization capabilities significantly improved situational awareness, reduced investigation time, and enhanced the ability of security analysts to understand attack patterns and system behavior within the monitored environment.

12.3 Performance Analysis

System performance was monitored throughout the testing period to assess the resource consumption of the monitoring platform under simulated attack conditions. The evaluation focused on CPU utilization, memory consumption, and disk I/O across all major components to determine the suitability of the architecture for continuous security monitoring.

Key performance metrics are summarized below:

Component	CPU Usage	Memory Usage	Disk I/O
Wazuh Manager	5-15%	512 MB	Low
Wazuh Indexer	10-25%	2-4 GB	Medium
Wazuh Dashboard	3-8%	512 MB	Low
Suricata IDS	8-20%	256-512 MB	Low

The results indicate that the system maintains stable performance under normal and moderate simulated attack loads. The Wazuh Indexer shows comparatively higher resource usage due to its role in storing, indexing, and enabling fast search queries across large volumes of security events. However, this usage remains within acceptable limits for a lab-scale deployment.

Alert generation latency, defined as the time between attack initiation and alert appearance on the dashboard, remained low across all tested scenarios. Network-based attacks such as port scanning and

ARP spoofing were typically detected and displayed within an average of under 10 seconds, while host-based detections such as file integrity monitoring events were observed within under 60 seconds due to periodic scanning intervals.

Overall, the performance evaluation demonstrates that the integrated Wazuh and Suricata architecture provides efficient resource utilization while maintaining near real-time detection capabilities. This confirms its suitability for laboratory environments, academic research setups, and small to medium enterprise Security Operations Center (SOC) deployments.

12.4 False Positive Analysis

False positive rates were assessed by monitoring the platform during normal operation without attack simulation. A small number of false positives were observed primarily from:

- Legitimate network scanning tools used by administrators generating low-severity scan alerts.
- Automated software update processes triggering file integrity monitoring alerts for legitimate package updates.
- DNS queries to newly registered domains triggering threat intelligence reputation alerts.

These false positives can be mitigated through rule tuning, trusted source whitelisting, and alert threshold adjustment. The overall false positive rate was considered acceptable for a laboratory environment using default configurations.

CHAPTER 13: ADVANTAGES

One of the most significant advantages of this monitoring solution is its zero licensing cost. Wazuh, Suricata, OpenSearch, Ubuntu, and Kali Linux are all open-source tools with permissive licenses that allow free use in both commercial and non-commercial environments. This makes it possible to deploy enterprise-grade security monitoring capabilities without the significant financial burden associated with commercial SIEM solutions, which can often exceed tens of thousands of dollars annually. In addition, the open-source nature allows organizations to inspect, modify, and customize the tools according to their specific security requirements.

The integrated Wazuh and Suricata platform provides a single pane of glass for security monitoring across endpoint and network dimensions. Security analysts can investigate alerts, analyze event timelines, and generate compliance reports from a unified dashboard interface, eliminating the need to switch between multiple security tools. This consolidation improves operational efficiency and reduces the likelihood of missing critical threat correlations across different layers of the infrastructure.

Near-real-time alert generation—typically within seconds of threat detection—enables rapid response to active security incidents. This is particularly important for detecting fast-moving threats such as ransomware encryption activities, brute force attacks, and lateral movement within networks, where even minor delays can significantly increase the potential impact and scope of compromise.

The Wazuh architecture also supports horizontal scalability, allowing the system to grow alongside organizational requirements. Additional Wazuh Indexer nodes can be introduced to enhance storage and search performance, while multiple Wazuh Manager instances can handle increasing numbers of agents and event volumes. This ensures that the platform remains effective in both small-scale lab environments and larger enterprise deployments without requiring a complete redesign.

Wazuh includes built-in compliance frameworks mapped to major regulatory standards such as PCI DSS, HIPAA, GDPR, and NIST 800-53. These frameworks enable organizations to use the platform not only for security monitoring but also for automated compliance reporting, reducing manual effort and simplifying audit preparation processes. This dual functionality enhances its value as both a security and governance tool.

Both Wazuh and Suricata benefit from large, active open-source communities that continuously contribute new features, detection rules, and documentation. Regular updates ensure that the platform evolves alongside emerging threats, providing ongoing improvements in detection accuracy and coverage. This community-driven development model reduces dependency on proprietary vendors while maintaining a high level of innovation and security effectiveness.

In addition to these benefits, the solution offers strong customization capabilities, allowing security teams to create custom detection rules, alerts, and dashboards tailored to specific organizational environments.

This flexibility ensures that the system can adapt to unique network architectures and evolving threat models.

The platform also enhances analyst productivity by reducing alert fatigue through rule tuning and severity-based filtering. By prioritizing high-confidence alerts and correlating related events, the system helps analysts focus on meaningful security incidents rather than being overwhelmed by low-priority noise.

Another important advantage of the proposed system is its strong suitability for educational and research purposes. Since the entire architecture is built using open-source tools, students and researchers can easily replicate the environment in virtual labs without requiring expensive infrastructure. This makes it highly effective for hands-on learning in areas such as SOC operations, intrusion detection, threat analysis, and incident response, while also allowing experimentation with attack simulations and defensive configurations in a safe and controlled environment.

Furthermore, the use of modular components improves maintainability and fault isolation. Each component—Wazuh Manager, Indexer, Dashboard, and Suricata—operates independently, making it easier to troubleshoot issues, apply updates, and upgrade individual services without disrupting the entire system.

CHAPTER 14: LIMITATIONS

Despite being an open-source solution, the integrated Wazuh platform has significant hardware resource requirements. The all-in-one deployment configuration requires a minimum of 8 GB RAM and 50 GB storage, which may be challenging for organizations with limited infrastructure budgets. As the number of monitored endpoints and event volume grows, hardware requirements scale proportionally, often necessitating additional indexing and storage optimization strategies to maintain performance.

The initial configuration of Wazuh and Suricata requires significant technical expertise. Proper tuning of detection rules, alert thresholds, active response scripts, and Suricata rule sets requires deep familiarity with both platforms. Misconfiguration can result in missed detections (false negatives) or alert flooding (false positives), both of which reduce the effectiveness of the monitoring solution and may overwhelm security analysts.

Security analysts new to SIEM and IDS technologies face a steep learning curve when working with the Wazuh platform. Effective threat investigation requires understanding of the Wazuh data model, OpenSearch query language, Suricata rule syntax, and general security analysis methodologies. As a result, organizations must invest in training and skill development to ensure analysts can fully utilize the system's capabilities.

The SOAR capabilities implemented in this project are limited to email notifications and Wazuh's built-in active response framework. Full SOAR functionality—such as automated ticket creation, multi-system orchestration, and advanced playbook automation—requires integration with dedicated SOAR platforms like TheHive, Shuffle, or Cortex XSOAR, which are beyond the scope of this project.

Suricata's detection capabilities are primarily signature-based, relying on predefined rule sets such as Emerging Threats to identify known attack patterns. This means that zero-day exploits or novel attack techniques without existing signatures may go undetected. Enhancing the system with behavioral analytics or machine learning-based anomaly detection would improve visibility into unknown or emerging threats.

In addition, log storage and retention can become a challenge in long-term deployments. High volumes of security events generated by continuous monitoring may lead to increased disk consumption, requiring log rotation policies, archiving strategies, or external storage solutions to manage historical data effectively.

Another limitation is alert noise in high-traffic environments. In networks with heavy legitimate traffic, the system may generate a large number of alerts, making it difficult for analysts to distinguish between genuine threats and benign activities without extensive rule tuning and filtering.

The system also has limited native support for full endpoint containment actions. While Wazuh provides some active response capabilities, advanced remediation actions such as automated device isolation,

network quarantine, or process termination at scale require additional third-party security orchestration tools.

Another limitation is the potential performance degradation under high event throughput conditions. When multiple attacks or large volumes of log data are generated simultaneously, components such as the Wazuh Indexer and Manager may experience increased CPU and memory usage, leading to slower alert processing and delayed dashboard updates. This can temporarily affect real-time visibility and reduce the speed at which analysts can investigate ongoing security incidents during peak load scenarios.

Finally, dependency on network visibility can be a constraint. Since Suricata relies on mirrored or tapped traffic, any misconfiguration in network tap/SPAN settings can result in incomplete traffic capture, leading to blind spots in detection coverage.

CHAPTER 15: FUTURE ENHANCEMENTS

Integration with a dedicated SOAR platform such as Shuffle (open-source) or TheHive would significantly enhance the automated response capabilities of the monitoring solution. A full SOAR integration would enable automated incident ticket creation, multi-tool orchestration workflows, and complex playbooks that can execute sophisticated response actions based on specific threat scenarios. This would also improve incident lifecycle management by ensuring structured tracking from detection to resolution.

Extending the notification system to include Telegram bot notifications, Slack webhooks, and Microsoft Teams integration would improve analyst awareness and response speed. Mobile push notifications would enable real-time alert delivery to analysts' mobile devices, ensuring visibility even when email access is limited, especially during on-call or off-site monitoring situations.

Wazuh's active response framework can be extended to automatically block attacker IP addresses at the firewall level when specific high-severity alerts are triggered. Integration with network firewalls and host-based firewalls (iptables/nftables) would enable automated threat containment, reducing the window of exposure for active attacks and improving overall incident response efficiency.

Integration with open-source threat intelligence feeds such as AbuseIPDB, VirusTotal, and MISP would enrich security alerts with reputation data, known malware associations, and threat actor attribution information. This contextual enrichment would significantly improve the efficiency of analyst investigations by prioritizing alerts from known malicious sources and reducing investigation time.

Incorporating machine learning capabilities into the monitoring platform would enable detection of sophisticated threats that evade signature-based detection. Anomaly detection models trained on baseline network and endpoint behavior could identify subtle deviations indicative of advanced persistent threats, insider threats, and zero-day exploits, improving detection coverage beyond traditional rule-based systems.

Migration of the Wazuh platform to a cloud environment (AWS, Azure, or GCP) would improve scalability, availability, and remote accessibility. Cloud deployment would also enable monitoring of cloud-native workloads and serverless functions, extending coverage to the full hybrid cloud infrastructure that modern organizations typically operate.

Expanding the laboratory to include multiple Wazuh agents monitoring diverse operating systems and application types would provide a more realistic representation of enterprise monitoring requirements. Adding Windows endpoints, web servers, database servers, and network devices as monitored targets

would validate the platform's cross-platform monitoring capabilities and improve the realism of simulation scenarios.

Implementation of a Security Data Lake architecture using scalable storage solutions such as OpenSearch cold storage or external object storage (e.g., S3-compatible systems) would significantly improve long-term log retention and historical threat analysis. This would allow organizations to store and analyze large volumes of security data without impacting primary system performance.

Enhancing endpoint security capabilities by integrating additional EDR-like features such as real-time process isolation, memory analysis, and advanced malware detection would strengthen host-level protection beyond basic log monitoring and file integrity checks.

Integration with identity and access management (IAM) systems such as Active Directory or LDAP would improve user-level visibility, enabling correlation between user authentication events and security incidents. This would enhance detection of compromised accounts and privilege misuse.

Deployment of deception technologies such as honeypots and honeytokens within the environment would help in early detection of attackers by attracting and identifying unauthorized activity in controlled decoy systems.

Improving visualization capabilities by integrating advanced analytics tools such as Kibana plugins or Grafana dashboards would provide more customizable and interactive threat intelligence views for analysts.

Development of automated incident response playbooks using scripting languages such as Python or Bash would enable faster remediation actions like service isolation, log collection, and forensic snapshot generation.

Enhancing network visibility by integrating NetFlow or Zeek (Bro) would provide deeper protocol-level analysis, improving detection of complex network behaviors such as lateral movement and data exfiltration attempts.

Introducing role-based access control (RBAC) improvements within the dashboard would allow better separation of duties between analysts, administrators, and auditors, improving operational security and compliance alignment.

Implementing containerized deployment using Docker or Kubernetes would improve system portability, simplify scaling, and make deployment across different environments more consistent and reproducible.

Finally, incorporating AI-driven alert prioritization and correlation would help reduce alert fatigue by automatically grouping related events, suppressing duplicate alerts, and highlighting the most critical threats based on behavioral patterns and historical context

CHAPTER 16: CONCLUSION

16.1 Project Summary

This project successfully designed and implemented an integrated cybersecurity monitoring laboratory using Wazuh SIEM/XDR, Suricata IDS, and Ubuntu Linux. The solution demonstrates comprehensive security monitoring capabilities including centralized log collection, real-time threat detection, automated alert generation, attack visibility, and email-based notification. The integration of endpoint and network monitoring provided a unified view of security events across the system, enabling effective correlation of threats from multiple sources.

The laboratory environment successfully detected all simulated attack scenarios, including Nmap reconnaissance scanning, TCP port scanning, and ARP spoofing attacks. Alerts were generated within seconds of attack initiation and displayed on the Wazuh Dashboard with full contextual information, enabling efficient investigation and rapid incident response. The system also demonstrated reliable log processing and consistent alert delivery across different attack types.

16.2 Key Achievements

The project achieved the following key outcomes:

- Successfully deployed a fully functional open-source SIEM/XDR platform using Wazuh and OpenSearch.
- Integrated Suricata IDS for network-level threat detection with seamless log forwarding to Wazuh.
- Simulated and detected real-world attack techniques including Nmap scanning, port scanning, and ARP spoofing.
- Implemented email-based automated alerting as a foundational SOAR capability.
- Demonstrated near-real-time alert generation with average detection latency under 10 seconds.
- Validated the cost-effectiveness of open-source security monitoring tools for laboratory and small enterprise environments.
- Established a scalable and modular architecture suitable for expansion into enterprise-level security monitoring environments
- Confirmed the feasibility of using open-source tools to replicate core SIEM and XDR functionalities without commercial licensing costs.

16.3 Broader Implications

This project demonstrates that effective cybersecurity monitoring is achievable without significant financial investment through the strategic use of open-source tools. Organizations of all sizes—including

small enterprises and academic institutions—can implement security monitoring capabilities comparable to commercial solutions by leveraging the Wazuh and Suricata ecosystem.

The increasing sophistication of cyber threats makes security monitoring an essential capability rather than an optional enhancement. Projects such as this contribute to improving overall cybersecurity readiness by providing practical, reproducible implementation models that can be adapted across different environments. Additionally, the integration of SIEM, IDS, and basic SOAR functionality highlights the importance of unified security architectures in modern IT infrastructure.

16.4 Conclusion

In conclusion, this project successfully demonstrates the design and implementation of a comprehensive, open-source cybersecurity monitoring laboratory. By integrating Wazuh SIEM, Suricata IDS, and automated alerting capabilities, the solution provides a practical platform for security monitoring, threat detection, and incident response suitable for academic, research, and small enterprise environments.

The project establishes a strong foundation for future enhancements such as full SOAR integration, machine learning-based anomaly detection, cloud-native deployment, and advanced threat intelligence integration. As cyber threats continue to evolve in complexity and scale, the adoption of integrated and automated monitoring solutions will remain critical for maintaining a strong and proactive security posture.

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APPENDIX I

CONFIGURATION AND IMPLEMENTATION DETAILS

Introduction

This appendix provides detailed information regarding the installation, configuration, and implementation procedures followed during the development of the cybersecurity monitoring laboratory. The purpose of this appendix is to document the technical configurations that enabled the successful integration of Wazuh SIEM, Suricata IDS, OpenSearch, Ubuntu Linux, and Kali Linux within a controlled virtualized environment. These implementation details serve as a reference for future deployments and ensure reproducibility of the project. The configuration process involved setting up virtual machines, installing monitoring components, establishing communication between agents and the manager, configuring intrusion detection rules, and implementing automated alert notification mechanisms.

Virtual Environment Configuration

The laboratory environment was created using Oracle VirtualBox virtualization software. Virtualization was selected because it provides a safe and isolated environment for conducting cybersecurity experiments without affecting production systems. Three virtual machines were deployed within the environment. The first machine hosted the Wazuh server and associated components, including the Wazuh Manager, OpenSearch Indexer, and Wazuh Dashboard. The second machine functioned as a monitored Ubuntu endpoint running the Wazuh agent. The third machine was configured with Kali Linux and served as the attacker system used to simulate cyberattacks.

Each virtual machine was allocated sufficient hardware resources to ensure stable operation. The Wazuh server received higher memory and processor allocations because it was responsible for processing logs, generating alerts, indexing data, and hosting the web dashboard. The Ubuntu endpoint and Kali Linux machines were configured with moderate resource allocations suitable for endpoint monitoring and attack simulation activities. All systems were connected through an isolated virtual network to facilitate secure communication and realistic attack scenarios.

Wazuh Installation and Configuration

The Wazuh platform was installed on an Ubuntu Server operating system. The installation process included deployment of the Wazuh Manager, OpenSearch Indexer, and Wazuh Dashboard components. The Wazuh Manager was configured to receive security events from monitored endpoints and perform rule-based analysis of incoming data. The OpenSearch Indexer provided centralized storage and indexing

capabilities, enabling efficient querying and retrieval of security events. The Wazuh Dashboard served as the visualization layer through which analysts could monitor alerts and investigate incidents.

Following installation, the Wazuh configuration file was modified to enable agent communication, alert generation, and email notifications. Custom settings were applied to define alert thresholds, log collection parameters, and active response options. The manager service was verified to ensure proper operation and communication with connected agents. Security monitoring policies were also configured to support file integrity monitoring, vulnerability detection, log analysis, and system auditing functions.

Agent Enrollment and Endpoint Monitoring

The Ubuntu endpoint was enrolled as a managed agent within the Wazuh environment. Agent registration was performed using enrollment keys generated by the Wazuh Manager. Once enrolled, the agent established secure communication with the server and began transmitting security events. Endpoint monitoring capabilities included system log collection, authentication monitoring, file integrity monitoring, process activity tracking, and vulnerability assessment.

The agent continuously collected security-relevant information from the operating system and transmitted the data to the Wazuh Manager for analysis. Successful enrollment was verified through the Wazuh Dashboard, where the agent appeared as an active monitored device. This configuration enabled centralized visibility into endpoint activities and allowed the system to detect suspicious behaviors occurring at the host level.

Suricata Installation and Integration

Suricata IDS was installed on the monitoring server to provide network-based intrusion detection capabilities. The installation process involved configuring the network interface for packet inspection and enabling the generation of EVE JSON output logs. Suricata was configured to utilize community-supported Emerging Threats rule sets, allowing detection of reconnaissance activities, attack signatures, malicious communications, and protocol anomalies.

The EVE JSON logging format was selected because it provides structured event data that can be easily parsed and analyzed. Wazuh was configured to ingest Suricata-generated logs, enabling centralized processing of network security events. This integration allowed network-based detections generated by Suricata to appear within the Wazuh Dashboard alongside host-based security events. The combined visibility significantly improved the effectiveness of threat detection and incident investigation activities.

Email Alert Configuration

An email notification mechanism was implemented to demonstrate basic Security Orchestration, Automation, and Response (SOAR) functionality. The Wazuh Manager was configured with SMTP server settings and alert notification rules. Email alerts were triggered whenever predefined security events exceeded specified severity thresholds. These notifications contained essential information regarding detected threats, including timestamps, source addresses, alert descriptions, and severity levels.

The email notification system improved security awareness by ensuring administrators received timely information regarding potentially malicious activities. Automated alert delivery reduced the need for continuous dashboard monitoring and demonstrated how security monitoring systems can support rapid incident response processes.

Summary

The implementation and configuration procedures described in this appendix established a functional cybersecurity monitoring laboratory capable of collecting, analyzing, and responding to security events. The integration of Wazuh, Suricata, OpenSearch, and automated alerting mechanisms provided comprehensive visibility across both endpoint and network layers. The resulting environment successfully supported the attack simulations and security monitoring objectives of the project.

APPENDIX II

SECURITY ANALYSIS AND THREAT MAPPING

Introduction

This appendix presents a detailed analysis of the security events generated during the implementation of the cybersecurity monitoring laboratory. The purpose of this appendix is to examine how various attack activities were identified, correlated, and classified within the monitoring environment. Security monitoring is not limited to generating alerts; it also involves understanding attacker behavior, analyzing threat indicators, and mapping activities to recognized cybersecurity frameworks. Through the integration of Wazuh and Suricata, the project provided visibility into both endpoint and network-level activities, enabling a more comprehensive understanding of attack patterns and security risks.

Threat Detection Methodology

The implemented monitoring environment utilized multiple detection mechanisms to identify suspicious activities. Wazuh analyzed operating system logs, authentication records, system events, and endpoint activities, while Suricata inspected network traffic for known attack signatures and anomalous communication patterns. Security events collected from these sources were processed through predefined detection rules, allowing the system to identify indicators of compromise associated with malicious activities. The combination of host-based and network-based monitoring significantly improved detection coverage and reduced the likelihood of undetected attacks.

MITRE ATT&CK Framework Mapping

To better understand attacker behavior, detected events were mapped to the MITRE ATT&CK Framework. This framework provides a standardized knowledge base of adversarial tactics and techniques based on real-world observations. Network reconnaissance activities performed using Nmap were mapped to the “Network Service Discovery” technique, while ARP spoofing attacks were associated with “Adversary-in-the-Middle” techniques. Mapping alerts to ATT&CK techniques helps analysts understand the objectives of attackers and supports more effective threat hunting and incident response activities.

The use of ATT&CK mapping also improves communication between security teams by providing a common language for describing attack behavior. Through this project, it was demonstrated that open-source monitoring platforms can support modern threat analysis methodologies commonly used within enterprise Security Operations Centers.

Security Risk Assessment

The identified threats were analyzed based on their likelihood of occurrence and potential impact on organizational assets. Reconnaissance activities were considered highly probable because they are frequently used during the initial stages of cyberattacks. ARP spoofing attacks were classified as high-impact threats due to their potential to facilitate credential theft, traffic interception, and unauthorized access to sensitive information.

The risk assessment process highlighted the importance of continuous monitoring and rapid incident detection. Early identification of suspicious activities can significantly reduce attacker dwell time and minimize the potential consequences of security incidents. The results obtained during the project demonstrate how integrated monitoring solutions contribute to improved organizational security posture.

Incident Investigation Process

When alerts were generated, investigators could use the Wazuh Dashboard to examine event details, review historical logs, and correlate related activities. This centralized approach simplified the investigation process and reduced the time required to determine the nature and severity of security events. The availability of both endpoint and network telemetry provided valuable context that supported accurate incident analysis and response planning.

Summary

The security analysis conducted during this project demonstrates the effectiveness of combining SIEM, XDR, and IDS technologies to detect and investigate cyber threats. The use of threat mapping frameworks, risk assessment methodologies, and centralized investigation tools provides a practical example of modern Security Operations Center practices and highlights the value of integrated security monitoring solutions.

APPENDIX III

EXPERIMENTAL RESULTS AND SUPPORTING EVIDENCE

Introduction

This appendix presents the results obtained during the testing and validation phases of the cybersecurity monitoring laboratory. The experiments were designed to evaluate the effectiveness of the implemented monitoring infrastructure in detecting and reporting simulated cyberattacks. Various attack scenarios were executed from the Kali Linux attacker machine while Wazuh and Suricata continuously monitored the environment. The resulting alerts, dashboard visualizations, and notification outputs provide evidence of successful threat detection and demonstrate the capabilities of the integrated security monitoring solution.

Nmap Reconnaissance Detection

One of the primary attack scenarios involved the use of Nmap to perform network reconnaissance and service discovery against monitored systems. Nmap is a widely used network scanning tool that allows attackers to identify active hosts, open ports, running services, and operating system information. During testing, several scanning techniques were executed, including TCP SYN scans, version detection scans, and aggressive service enumeration scans.

Suricata successfully identified scanning behavior using signature-based detection rules. Generated alerts contained information regarding the scanning source, targeted systems, and associated attack signatures. These alerts were forwarded to Wazuh for centralized analysis and visualization. The detection results confirmed that the monitoring infrastructure could identify reconnaissance activities commonly associated with the early stages of cyberattacks.

ARP Spoofing Attack Detection

ARP spoofing was performed to simulate a Man-in-the-Middle attack scenario. The attacker machine transmitted forged Address Resolution Protocol messages in an attempt to manipulate network communications between victim systems and the gateway. This attack demonstrates how adversaries may intercept, modify, or monitor network traffic without authorization.

The monitoring environment successfully detected anomalous network behavior associated with ARP spoofing activities. Network traffic analysis revealed suspicious ARP packet patterns that differed from normal communication behavior. Security alerts generated by the system provided visibility into the attack and demonstrated the effectiveness of network monitoring in identifying local network threats.

Dashboard Visualization and Alert Analysis

The Wazuh Dashboard served as the primary interface for analyzing security events generated during attack simulations. Real-time dashboards displayed event summaries, agent status information, alert severity distributions, and detailed security logs. Analysts could filter and investigate alerts based on source IP addresses, event categories, timestamps, and rule identifiers.

Visualization capabilities significantly improved situational awareness by presenting security information in an organized and accessible format. Dashboard widgets enabled rapid identification of suspicious activities and supported incident investigation processes. The integration of network and endpoint security events provided valuable context for understanding attack progression and potential impacts.

Email Notification Results

Automated email alerts were generated when predefined detection rules were triggered. These notifications included detailed information regarding detected attacks and served as evidence of successful alert automation. Email delivery occurred shortly after attack detection, demonstrating the ability of the monitoring system to provide timely notifications to administrators.

The implementation of automated notifications illustrates a foundational SOAR capability. By reducing reliance on manual monitoring, email alerts improve incident awareness and support faster response times. The results confirmed that the configured notification mechanism functioned reliably throughout testing.

Performance Evaluation

The overall performance of the monitoring environment was assessed by observing detection speed, alert generation efficiency, system responsiveness, and resource utilization. Both Wazuh and Suricata operated reliably under simulated attack conditions without significant degradation in performance. Security events were processed and visualized within seconds of occurrence, providing near real-time visibility into attack activities.

The integration of endpoint and network monitoring technologies improved detection coverage and reduced blind spots. Combining host-based and network-based security data enabled more comprehensive analysis than either technology could provide independently. These findings demonstrate the value of adopting a layered monitoring approach within modern cybersecurity environments.

Summary

The experimental results presented in this appendix validate the effectiveness of the implemented cybersecurity monitoring laboratory. Attack simulations generated meaningful security events that were successfully detected, analyzed, and reported through the integrated Wazuh and Suricata platform. The

findings demonstrate the practical benefits of combining SIEM, XDR, IDS, and SOAR concepts within a unified monitoring framework and highlight the suitability of open-source technologies for cybersecurity education, research, and operational security monitoring.

APPENDIX IV

FUTURE RESEARCH DIRECTIONS AND PROJECT EXTENSIONS

Introduction

This appendix discusses potential enhancements and future research opportunities related to the cybersecurity monitoring laboratory. Although the implemented solution successfully demonstrated core SIEM, XDR, IDS, and SOAR functionalities, cybersecurity is a continuously evolving field that requires ongoing adaptation to emerging threats and technological advancements. The laboratory environment developed during this project provides a strong foundation for future experimentation, research, and system expansion.

Advanced Threat Intelligence Integration

One significant enhancement involves integrating external threat intelligence feeds into the monitoring platform. Threat intelligence provides information regarding malicious IP addresses, domains, file hashes, and indicators of compromise collected from global cybersecurity communities. By incorporating threat intelligence sources, the monitoring system can identify emerging threats more effectively and improve detection accuracy. Future implementations could automate the ingestion of threat intelligence data and correlate it with locally observed security events.

Machine Learning and Artificial Intelligence

The application of machine learning and artificial intelligence represents another promising area for future research. Traditional detection methods rely heavily on predefined rules and signatures, which may not identify previously unknown threats. Machine learning algorithms can analyze large volumes of security data, establish behavioral baselines, and detect anomalies that may indicate malicious activity. Future work could investigate the integration of AI-based detection techniques with Wazuh and Suricata to improve threat identification and reduce false positives.

Cloud Security Monitoring

As organizations increasingly adopt cloud computing platforms, security monitoring must extend beyond traditional on-premises environments. Future versions of the laboratory could incorporate monitoring capabilities for cloud platforms such as AWS, Microsoft Azure, and Google Cloud Platform. Cloud monitoring would provide visibility into virtual machines, storage services, identity systems, and cloud-native applications, reflecting the requirements of modern enterprise environments.

Advanced SOAR Capabilities

The current implementation demonstrates basic automation through email-based alert notifications. Future enhancements could include advanced SOAR functionality such as automated ticket creation, endpoint isolation, firewall rule updates, malicious IP blocking, and integration with incident response

platforms. These capabilities would enable faster containment of security incidents and further reduce the workload placed on security analysts.

Threat Hunting and Security Analytics

Future research may also explore proactive threat hunting methodologies. Rather than waiting for alerts to occur, threat hunting involves actively searching for hidden adversaries within monitored environments. The integration of advanced analytics, behavioral monitoring, and threat intelligence could support more sophisticated threat hunting activities and improve overall security effectiveness.

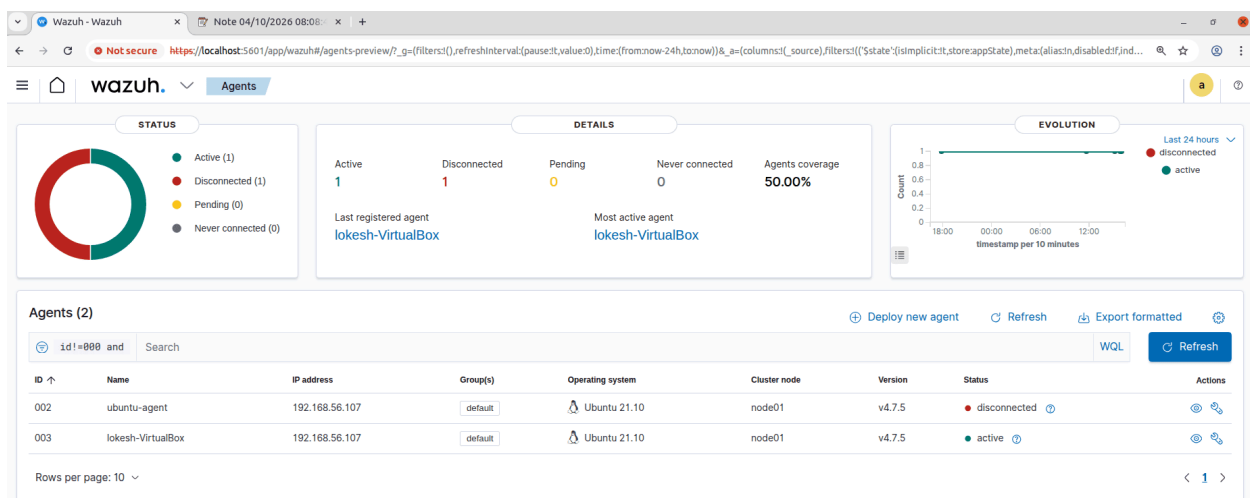
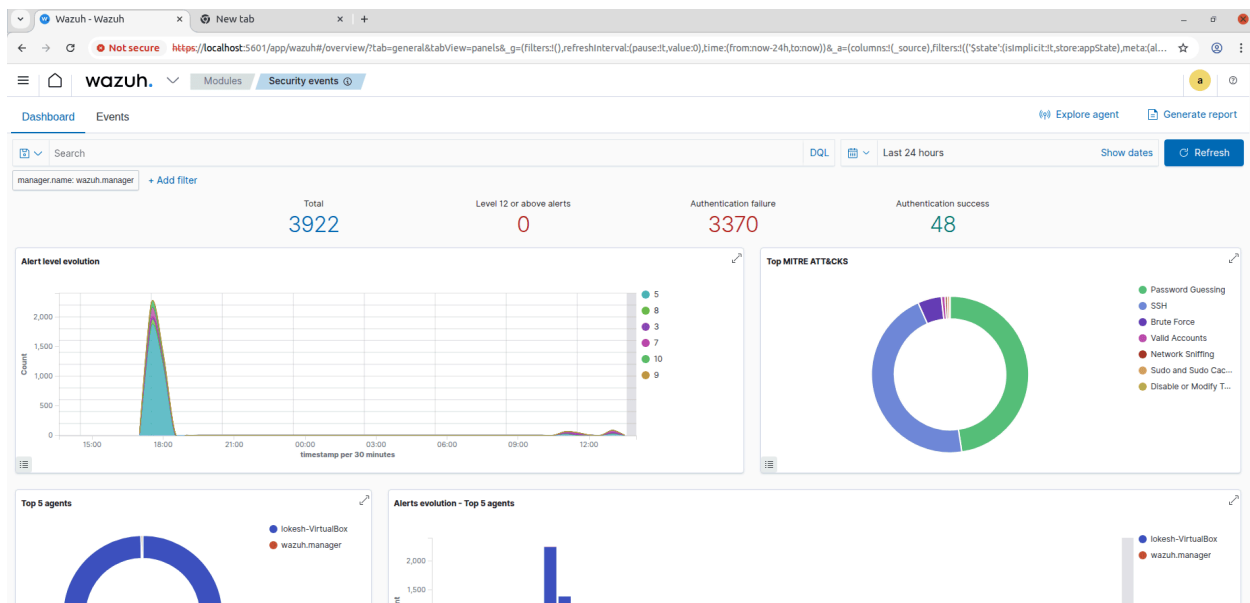
Scalability and Enterprise Deployment

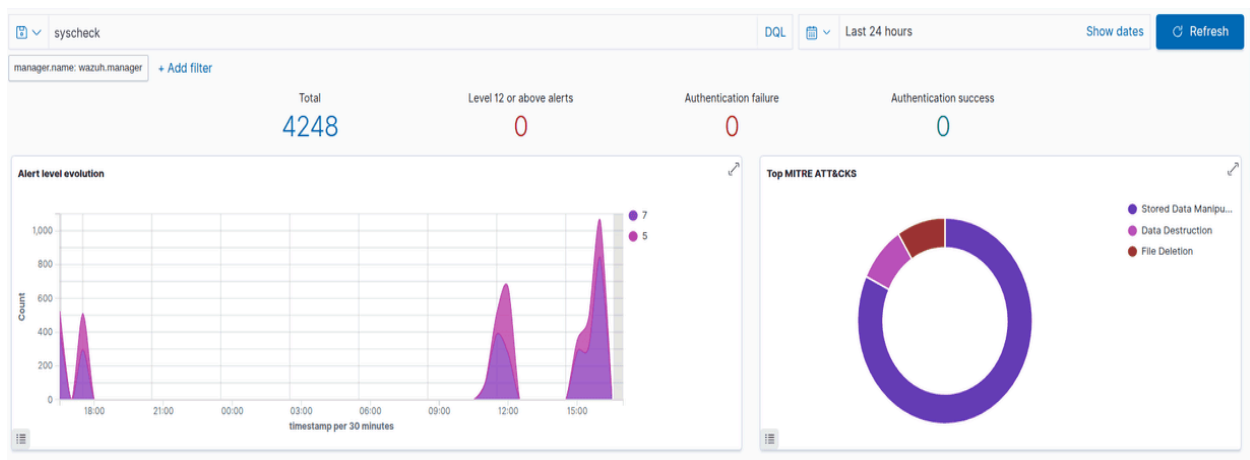
The current laboratory environment was designed primarily for educational and research purposes. Future work could investigate large-scale deployment architectures capable of supporting hundreds or thousands of monitored systems. Areas of interest include distributed Wazuh architectures, high-availability configurations, load balancing mechanisms, and large-scale log management strategies. Such enhancements would improve system resilience and support enterprise-level security monitoring requirements.

Summary

The future enhancements described in this appendix demonstrate the scalability and adaptability of the implemented cybersecurity monitoring laboratory. Emerging technologies such as artificial intelligence, cloud security monitoring, threat intelligence integration, and advanced automation provide numerous opportunities for expanding the capabilities of the platform. These developments would further align the laboratory with modern cybersecurity practices and support ongoing research, education, and innovation in the field of security monitoring.

SCREENSHOTS





Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Apr 15, 2026 @ 16:46:08.201	003	lokesh-VirtualBox	T1070.004 T1485	Defense Evasion, Impact	File deleted.	7	553
> Apr 15, 2026 @ 16:46:08.093	003	lokesh-VirtualBox	T1070.004 T1485	Defense Evasion, Impact	File deleted.	7	553
> Apr 15, 2026 @ 16:45:02.104	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550
> Apr 15, 2026 @ 16:45:01.816	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550
> Apr 15, 2026 @ 16:45:01.291	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550
> Apr 15, 2026 @ 16:45:01.276	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550
> Apr 15, 2026 @ 16:45:01.245	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550
> Apr 15, 2026 @ 16:45:01.176	003	lokesh-VirtualBox	T1565.001	Impact	Integrity checksum changed.	7	550

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148:20.783+0000","rule":{"level":5,"description":"sshd: Attempt to login using a non-existent user","id":"5710","mitre":{"id":["T1110.001","T1021.004"],"tactic":["Credential Access","Lateral Movement"],"technique":["Password Guessing","SSH"]},"firedtmes":2,"mail":false,"groups":["syslog","sshd"],"authentication_failed","invalid_login"},"gdpri":["IV.35.7.d","IV.32.2"],"gpg13":["7.1"],"hipaa":["164.312.b"],"nist_800_53":["AU.14","AC.7","AU.6"],"pci_dss":["10.2.4","10.2.5","10.6.1"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"003","name":"lokesh-VirtualBox","ip":"192.168.56.107"},"manager":{"name":"wazuh.manager"},"id":"1776239706.70793","full_log":"Apr 15 12:18:19 lokesh-VirtualBox sshd[14827]: Disconnected from invalid user wronguser 192.168.56.108 port 33154 [preauth]","predecoder":{"program_name":"sshd","timestamp":"Apr 15 12:18:19","hostname":"lokesh-VirtualBox"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.56.108","srcport":"33154","srcuser":"wronguser"},"location":"/var/log/auth.log"},"program":"active-response/bin/firewall-drop"}}

2026/04/15 12:18:21 active-response/bin/firewall-drop: Ended
2026/04/15 13:25:19 active-response/bin/firewall-drop: Starting
2026/04/15 13:25:19 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":[],"alert":{"timestamp":"2026-04-15T07:55:18.655+0000","rule":{"level":5,"description":"sshd: Attempt to login using a non-existent user","id":"5710","mitre":{"id":["T1110.001","T1021.004"],"tactic":["Credential Access","Lateral Movement"],"technique":["Password Guessing","SSH"]},"firedtmes":2,"mail":false,"groups":["syslog","sshd"],"authentication_failed","invalid_login"},"gdpri":["IV.35.7.d","IV.32.2"],"gpg13":["7.1"],"hipaa":["164.312.b"],"nist_800_53":["AU.14","AC.7","AU.6"],"pci_dss":["10.2.4","10.2.5","10.6.1"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"003","name":"lokesh-VirtualBox","ip":"192.168.56.107"},"manager":{"name":"wazuh.manager"},"id":"1776239718.119472","full_log":"Apr 15 13:25:18 lokesh-VirtualBox sshd[8579]: Invalid user lokibhai from 192.168.56.108 port 36764","predecoder":{"program_name":"sshd","timestamp":"Apr 15 13:25:18","hostname":"lokesh-VirtualBox"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.56.108","srcport":"36764","srcuser":"lokibhai"},"location":"/var/log/auth.log"},"program":"active-response/bin/firewall-drop"}}

2026/04/15 13:25:19 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"firewall-drop","module":"active-response"},"command":"check_keys","parameters":{"keys":["192.168.56.108"]}}
2026/04/15 13:25:19 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"continue","parameters":{"extra_args":[],"alert":{"timestamp":"2026-04-15T07:55:18.655+0000","rule":{"level":5,"description":"sshd: Attempt to login using a non-existent user","id":"5710","mitre":{"id":["T1110.001","T1021.004"],"tactic":["Credential Access","Lateral Movement"],"technique":["Password Guessing","SSH"]},"firedtmes":2,"mail":false,"groups":["syslog","sshd"],"authentication_failed","invalid_login"},"gdpri":["IV.35.7.d","IV.32.2"],"gpg13":["7.1"],"hipaa":["164.312.b"],"nist_800_53":["AU.14","AC.7","AU.6"],"pci_dss":["10.2.4","10.2.5","10.6.1"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"003","name":"lokesh-VirtualBox","ip":"192.168.56.107"},"manager":{"name":"wazuh.manager"},"id":"1776239718.119472","full_log":"Apr 15 13:25:18 lokesh-VirtualBox sshd[8579]: Invalid user lokibhai from 192.168.56.108 port 36764","predecoder":{"program_name":"sshd","timestamp":"Apr 15 13:25:18","hostname":"lokesh-VirtualBox"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.56.108","srcport":"36764","srcuser":"lokibhai"},"location":"/var/log/auth.log"},"program":"active-response/bin/firewall-drop"}}

2026/04/15 13:25:20 active-response/bin/firewall-drop: Ended
2026/04/15 13:25:20 active-response/bin/firewall-drop: Starting
2026/04/15 13:25:20 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":[],"alert":{"timestamp":"2026-04-15T07:55:20.461+0000","rule":{"level":5,"description":"sshd: Attempt to login using a non-existent user","id":"5710","mitre":{"id":["T1110.001","T1021.004"],"tactic":["Credential Access","Lateral Movement"],"technique":["Password Guessing","SSH"]},"firedtmes":2,"mail":false,"groups":["syslog","sshd"],"authentication_failed","invalid_login"},"gdpri":["IV.35.7.d","IV.32.2"],"gpg13":["7.1"],"hipaa":["164.312.b"],"nist_800_53":["AU.14","AC.7","AU.6"],"pci_dss":["10.2.4","10.2.5","10.6.1"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"003","name":"lokesh-VirtualBox","ip":"192.168.56.107"},"manager":{"name":"wazuh.manager"},"id":"1776239720.120020","full_log":"Apr 15 13:25:19 lokesh-VirtualBox sshd[8579]: Disconnected from invalid user lokibhai 192.168.56.108 port 36764 [preauth]","predecoder":{"program_name":"sshd","timestamp":"Apr 15 13:25:19","hostname":"lokesh-VirtualBox"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.56.108","srcport":"36764","srcuser":"lokibhai"},"location":"/var/log/auth.log"},"program":"active-response/bin/firewall-drop"}}

2026/04/15 13:25:20 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"firewall-drop","module":"active-response"},"command":"check_keys","parameters":{"keys":["192.168.56.108"]}}
2026/04/15 13:25:20 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"continue","parameters":{"extra_args":[],"alert":{"timestamp":"2026-04-15T07:55:20.461+0000","rule":{"level":5,"description":"sshd: Attempt to login using a non-existent user","id":"5710","mitre":{"id":["T1110.001","T1021.004"],"tactic":["Credential Access","Lateral Movement"],"technique":["Password Guessing","SSH"]},"firedtmes":2,"mail":false,"groups":["syslog","sshd"],"authentication_failed","invalid_login"},"gdpri":["IV.35.7.d","IV.32.2"],"gpg13":["7.1"],"hipaa":["164.312.b"],"nist_800_53":["AU.14","AC.7","AU.6"],"pci_dss":["10.2.4","10.2.5","10.6.1"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"003","name":"lokesh-VirtualBox","ip":"192.168.56.107"},"manager":{"name":"wazuh.manager"},"id":"1776239720.120020","full_log":"Apr 15 13:25:19 lokesh-VirtualBox sshd[8579]: Disconnected from invalid user lokibhai 192.168.56.108 port 36764 [preauth]","predecoder":{"program_name":"sshd","timestamp":"Apr 15 13:25:19","hostname":"lokesh-VirtualBox"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.56.108","srcport":"36764","srcuser":"lokibhai"},"location":"/var/log/auth.log"},"program":"active-response/bin/firewall-drop"}}

2026/04/15 13:25:20 active-response/bin/firewall-drop: Ended

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